

resistance $1.2\ \Omega$. An aluminum cup with mass $0.130\ \text{kg}$ contains $0.200\ \text{kg}$ of water. The heating element is placed in the water and the electrical energy dissipated in the resistance of the heating element all goes into the cup and water. The element itself has very small mass. How much time does it take for the temperature of the cup and water to rise from 21.2°C to 34.5°C ? (The change of the resistance of the Nichrome due to its temperature change can be neglected.)

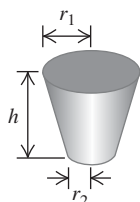
25.62 •• A resistor with resistance R is connected to a battery that has emf $12.0\ \text{V}$ and internal resistance $r = 0.40\ \Omega$. For what two values of R will the power dissipated in the resistor be $80.0\ \text{W}$?

25.63 •• CP BIO Struck by Lightning. Lightning strikes can involve currents as high as $25,000\ \text{A}$ that last for about $40\ \mu\text{s}$. If a person is struck by a bolt of lightning with these properties, the current will pass through his body. We shall assume that his mass is $75\ \text{kg}$, that he is wet (after all, he is in a rainstorm) and therefore has a resistance of $1.0\ \text{k}\Omega$, and that his body is all water (which is reasonable for a rough, but plausible, approximation). (a) By how many degrees Celsius would this lightning bolt increase the temperature of $75\ \text{kg}$ of water? (b) Given that the internal body temperature is about 37°C , would the person's temperature actually increase that much? Why not? What would happen first?

25.64 •• In the Bohr model of the hydrogen atom, the electron makes $6.0 \times 10^{15}\ \text{rev/s}$ around the nucleus. What is the average current at a point on the orbit of the electron?

25.65 • CALC A material of resistivity ρ is formed into a solid, truncated cone of height h and radii r_1 and r_2 at either end (Fig. P25.65). (a) Calculate the resistance of the cone between the two flat end faces. (*Hint:* Imagine slicing the cone into very many thin disks, and calculate the resistance of one such disk.) (b) Show that your result agrees with Eq. (25.10) when $r_1 = r_2$.

Figure P25.65



25.66 • CALC The region between two concentric conducting spheres with radii a and b is filled with a conducting material with resistivity ρ . (a) Show that the resistance between the spheres is given by

$$R = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

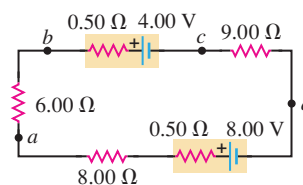
(b) Derive an expression for the current density as a function of radius, in terms of the potential difference V_{ab} between the spheres. (c) Show that the result in part (a) reduces to Eq. (25.10) when the separation $L = b - a$ between the spheres is small.

25.67 •• The temperature coefficient of resistance α in Eq. (25.12) equals the temperature coefficient of resistivity α in Eq. (25.6) only if the coefficient of thermal expansion is small. A cylindrical column of mercury is in a vertical glass tube. At 20°C , the length of the mercury column is $12.0\ \text{cm}$. The diameter of the mercury column is $1.6\ \text{mm}$ and doesn't change with temperature because glass has a small coefficient of thermal expansion. The coefficient of volume expansion of the mercury is given in Table 17.2, its resistivity at 20°C is given in Table 25.1, and its temperature coefficient of resistivity is given in Table 25.2. (a) At 20°C , what is the resistance between the ends of the mercury column? (b) The mercury column is heated to 60°C . What is the change in its resistivity? (c) What is the change in its length? Explain why the coefficient of volume expansion, rather than the coefficient of linear expansion, determines the change in length. (d) What is the change in its resistance? (*Hint:* Since the percentage changes in ρ and L are small, you may find it helpful to derive from Eq. (25.10) an

equation for ΔR in terms of $\Delta\rho$ and ΔL .) (e) What is the temperature coefficient of resistance α for the mercury column, as defined in Eq. (25.12)? How does this value compare with the temperature coefficient of resistivity? Is the effect of the change in length important?

25.68 • (a) What is the potential difference V_{ad} in the circuit of Fig. P25.68? (b) What is the terminal voltage of the 4.00-V battery? (c) A battery with emf $10.30\ \text{V}$ and internal resistance $0.50\ \Omega$ is inserted in the circuit at d , with its negative terminal connected to the negative terminal of the 8.00-V battery. What is the difference of potential V_{bc} between the terminals of the 4.00-V battery now?

Figure P25.68



25.69 • The potential difference across the terminals of a battery is $8.40\ \text{V}$ when there is a current of $1.50\ \text{A}$ in the battery from the negative to the positive terminal. When the current is $3.50\ \text{A}$ in the reverse direction, the potential difference becomes $10.20\ \text{V}$. (a) What is the internal resistance of the battery? (b) What is the emf of the battery?

25.70 •• BIO A person with body resistance between his hands of $10\ \text{k}\Omega$ accidentally grasps the terminals of a 14-kV power supply. (a) If the internal resistance of the power supply is $2000\ \Omega$, what is the current through the person's body? (b) What is the power dissipated in his body? (c) If the power supply is to be made safe by increasing its internal resistance, what should the internal resistance be for the maximum current in the above situation to be $1.00\ \text{mA}$ or less?

25.71 • BIO The average bulk resistivity of the human body (apart from surface resistance of the skin) is about $5.0\ \Omega \cdot \text{m}$. The conducting path between the hands can be represented approximately as a cylinder $1.6\ \text{m}$ long and $0.10\ \text{m}$ in diameter. The skin resistance can be made negligible by soaking the hands in salt water. (a) What is the resistance between the hands if the skin resistance is negligible? (b) What potential difference between the hands is needed for a lethal shock current of $100\ \text{mA}$? (Note that your result shows that small potential differences produce dangerous currents when the skin is damp.) (c) With the current in part (b), what power is dissipated in the body?

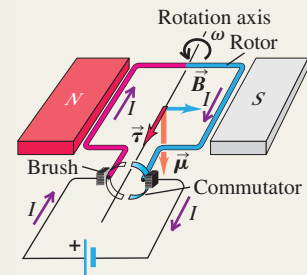
25.72 • A typical cost for electric power is $\$0.120$ per kilowatt-hour. (a) Some people leave their porch light on all the time. What is the yearly cost to keep a 75-W bulb burning day and night? (b) Suppose your refrigerator uses $400\ \text{W}$ of power when it's running, and it runs 8 hours a day. What is the yearly cost of operating your refrigerator?

25.73 •• A 12.6-V car battery with negligible internal resistance is connected to a series combination of a $3.2\text{-}\Omega$ resistor that obeys Ohm's law and a thermistor that does not obey Ohm's law but instead has a current-voltage relationship $V = \alpha I + \beta I^2$, with $\alpha = 3.8\ \Omega$ and $\beta = 1.3\ \Omega/\text{A}$. What is the current through the $3.2\text{-}\Omega$ resistor?

25.74 •• A cylindrical copper cable $1.50\ \text{km}$ long is connected across a 220.0-V potential difference. (a) What should be its diameter so that it produces heat at a rate of $75.0\ \text{W}$? (b) What is the electric field inside the cable under these conditions?

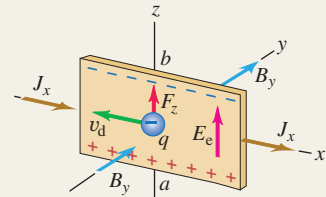
25.75 •• A Nonideal Ammeter. Unlike the idealized ammeter described in Section 25.4, any real ammeter has a nonzero resistance. (a) An ammeter with resistance R_A is connected in series with a resistor R and a battery of emf $\mathcal{E}$ and internal resistance r . The current measured by the ammeter is I_A . Find the current

Electric motors: In a dc motor a magnetic field exerts a torque on a current in the rotor. Motion of the rotor through the magnetic field causes an induced emf called a back emf. For a series motor, in which the rotor coil is in parallel with coils that produce the magnetic field, the terminal voltage is the sum of the back emf and the drop Ir across the internal resistance. (See Example 27.11.)



The Hall effect: The Hall effect is a potential difference perpendicular to the direction of current in a conductor, when the conductor is placed in a magnetic field. The Hall potential is determined by the requirement that the associated electric field must just balance the magnetic force on a moving charge. Hall-effect measurements can be used to determine the sign of charge carriers and their concentration n . (See Example 27.12.)

$$nq = \frac{-J_x B_y}{E_z} \quad (27.30)$$



BRIDGING PROBLEM

Magnetic Torque on a Current-Carrying Ring

A circular ring with area 4.45 cm^2 is carrying a current of 12.5 A . The ring, initially at rest, is immersed in a region of uniform magnetic field given by $\vec{B} = (1.15 \times 10^{-2} \text{ T})(12\hat{i} + 3\hat{j} - 4\hat{k})$. The ring is positioned initially such that its magnetic moment is given by $\vec{\mu}_i = \mu(-0.800\hat{i} + 0.600\hat{j})$, where μ is the (positive) magnitude of the magnetic moment. (a) Find the initial magnetic torque on the ring. (b) The ring (which is free to rotate around one diameter) is released and turns through an angle of 90.0° , at which point its magnetic moment is given by $\vec{\mu}_f = -\mu\hat{k}$. Determine the decrease in potential energy. (c) If the moment of inertia of the ring about a diameter is $8.50 \times 10^{-7} \text{ kg} \cdot \text{m}^2$, determine the angular speed of the ring as it passes through the second position.

SOLUTION GUIDE

See MasteringPhysics® study area for a Video Tutor solution.



IDENTIFY and SET UP

1. The current-carrying ring acts as a magnetic dipole, so you can use the equations for a magnetic dipole in a uniform magnetic field.

2. There are no nonconservative forces acting on the ring as it rotates, so the sum of its rotational kinetic energy (discussed in Section 9.4) and the potential energy is conserved.

EXECUTE

3. Use the vector expression for the torque on a magnetic dipole to find the answer to part (a). (*Hint:* You may want to review Section 1.10.)
4. Find the change in potential energy from the first orientation of the ring to the second orientation.
5. Use your result from step 4 to find the rotational kinetic energy of the ring when it is in the second orientation.
6. Use your result from step 5 to find the ring's angular speed when it is in the second orientation.

EVALUATE

7. If the ring were free to rotate around *any* diameter, in what direction would the magnetic moment point when the ring is in a state of stable equilibrium?

Problems

For instructor-assigned homework, go to www.masteringphysics.com



•, ••, •••: Problems of increasing difficulty. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q27.1 Can a charged particle move through a magnetic field without experiencing any force? If so, how? If not, why not?

Q27.2 At any point in space, the electric field $\vec{E}$ is defined to be in the direction of the electric force on a positively charged particle at that point. Why don't we similarly define the magnetic field $\vec{B}$ to

be in the direction of the magnetic force on a moving, positively charged particle?

Q27.3 Section 27.2 describes a procedure for finding the direction of the magnetic force using your right hand. If you use the same procedure, but with your left hand, will you get the correct direction for the force? Explain.

Q27.4 The magnetic force on a moving charged particle is always perpendicular to the magnetic field $\vec{B}$. Is the trajectory of a moving charged particle always perpendicular to the magnetic field lines? Explain your reasoning.

Q27.5 A charged particle is fired into a cubical region of space where there is a uniform magnetic field. Outside this region, there is no magnetic field. Is it possible that the particle will remain inside the cubical region? Why or why not?

Q27.6 If the magnetic force does no work on a charged particle, how can it have any effect on the particle's motion? Are there other examples of forces that do no work but have a significant effect on a particle's motion?

Q27.7 A charged particle moves through a region of space with constant velocity (magnitude and direction). If the external magnetic field is zero in this region, can you conclude that the external electric field in the region is also zero? Explain. (By "external" we mean fields other than those produced by the charged particle.) If the external electric field is zero in the region, can you conclude that the external magnetic field in the region is also zero?

Q27.8 How might a loop of wire carrying a current be used as a compass? Could such a compass distinguish between north and south? Why or why not?

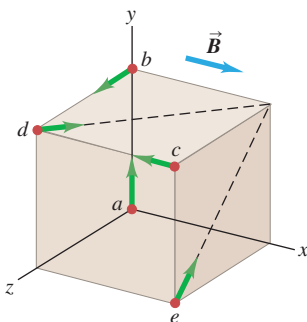
Q27.9 How could the direction of a magnetic field be determined by making only *qualitative* observations of the magnetic force on a straight wire carrying a current?

Q27.10 A loose, floppy loop of wire is carrying current I . The loop of wire is placed on a horizontal table in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the table. This causes the loop of wire to expand into a circular shape while still lying on the table. In a diagram, show all possible orientations of the current I and magnetic field $\vec{B}$ that could cause this to occur. Explain your reasoning.

Q27.11 Several charges enter a uniform magnetic field directed into the page. (a) What path would a positive charge q moving with a velocity of magnitude v follow through the field? (b) What path would a positive charge q moving with a velocity of magnitude $2v$ follow through the field? (c) What path would a negative charge $-q$ moving with a velocity of magnitude v follow through the field? (d) What path would a neutral particle follow through the field?

Q27.12 Each of the lettered points at the corners of the cube in Fig. Q27.12 represents a positive charge q moving with a velocity of magnitude v in the direction indicated. The region in the figure is in a uniform magnetic field $\vec{B}$, parallel to the x -axis and directed toward the right. Which charges experience a force due to $\vec{B}$? What is the direction of the force on each charge?

Figure Q27.12



Q27.13 A student claims that if lightning strikes a metal flagpole, the force exerted by the earth's magnetic field on the current in the pole can be large enough to bend it. Typical lightning currents are of the order of 10^4 to 10^5 A. Is the student's opinion justified? Explain your reasoning.

Q27.14 Could an accelerator be built in which *all* the forces on the particles, for steering and for increasing speed, are magnetic forces? Why or why not?

Q27.15 An ordinary loudspeaker such as that shown in Fig. 27.28 should not be placed next to a computer monitor or TV screen. Why not?

Q27.16 The magnetic force acting on a charged particle can never do work because at every instant the force is perpendicular to the velocity. The torque exerted by a magnetic field can do work on a current loop when the loop rotates. Explain how these seemingly contradictory statements can be reconciled.

Q27.17 If an emf is produced in a dc motor, would it be possible to use the motor somehow as a generator or source, taking power out of it rather than putting power into it? How might this be done?

Q27.18 When the polarity of the voltage applied to a dc motor is reversed, the direction of motion does *not* reverse. Why not? How *could* the direction of motion be reversed?

Q27.19 In a Hall-effect experiment, is it possible that *no* transverse potential difference will be observed? Under what circumstances might this happen?

Q27.20 Hall-effect voltages are much greater for relatively poor conductors (such as germanium) than for good conductors (such as copper), for comparable currents, fields, and dimensions. Why?

EXERCISES

Section 27.2 Magnetic Field

27.1 • A particle with a charge of -1.24×10^{-8} C is moving with instantaneous velocity $\vec{v} = (4.19 \times 10^4 \text{ m/s})\hat{i} + (-3.85 \times 10^4 \text{ m/s})\hat{j}$. What is the force exerted on this particle by a magnetic field (a) $\vec{B} = (1.40 \text{ T})\hat{i}$ and (b) $\vec{B} = (1.40 \text{ T})\hat{k}$?

27.2 • A particle of mass 0.195 g carries a charge of -2.50×10^{-8} C. The particle is given an initial horizontal velocity that is due north and has magnitude 4.00×10^4 m/s. What are the magnitude and direction of the minimum magnetic field that will keep the particle moving in the earth's gravitational field in the same horizontal, northward direction?

27.3 • In a 1.25-T magnetic field directed vertically upward, a particle having a charge of magnitude $8.50 \mu\text{C}$ and initially moving northward at 4.75 km/s is deflected toward the east. (a) What is the sign of the charge of this particle? Make a sketch to illustrate how you found your answer. (b) Find the magnetic force on the particle.

27.4 • A particle with mass 1.81×10^{-3} kg and a charge of 1.22×10^{-8} C has, at a given instant, a velocity $\vec{v} = (3.00 \times 10^4 \text{ m/s})\hat{j}$. What are the magnitude and direction of the particle's acceleration produced by a uniform magnetic field $\vec{B} = (1.63 \text{ T})\hat{i} + (0.980 \text{ T})\hat{j}$?

27.5 • An electron experiences a magnetic force of magnitude 4.60×10^{-15} N when moving at an angle of 60.0° with respect to a magnetic field of magnitude 3.50×10^{-3} T. Find the speed of the electron.

27.6 • An electron moves at 2.50×10^6 m/s through a region in which there is a magnetic field of unspecified direction and magnitude 7.40×10^{-2} T. (a) What are the largest and smallest possible magnitudes of the acceleration of the electron due to the magnetic field? (b) If the actual acceleration of the electron is one-fourth of the largest magnitude in part (a), what is the angle between the electron velocity and the magnetic field?

27.7 •• CP A particle with charge $7.80 \mu\text{C}$ is moving with velocity $\vec{v} = -(3.80 \times 10^3 \text{ m/s})\hat{j}$. The magnetic force on the particle is measured to be $\vec{F} = +(7.60 \times 10^{-3} \text{ N})\hat{i} - (5.20 \times 10^{-3} \text{ N})\hat{k}$. (a) Calculate all the components of the magnetic field you can from this information. (b) Are there components of the magnetic field that are not determined by the measurement of the force? Explain. (c) Calculate the scalar product $\vec{B} \cdot \vec{F}$. What is the angle between $\vec{B}$ and $\vec{F}$?

27.8 •• CP A particle with charge -5.60 nC is moving in a uniform magnetic field $\vec{B} = -(1.25 \text{ T})\hat{k}$. The magnetic force on the

particle is measured to be $\vec{F} = -(3.40 \times 10^{-7} \text{ N})\hat{i} + (7.40 \times 10^{-7} \text{ N})\hat{j}$. (a) Calculate all the components of the velocity of the particle that you can from this information. (b) Are there components of the velocity that are not determined by the measurement of the force? Explain. (c) Calculate the scalar product $\vec{v} \cdot \vec{F}$. What is the angle between $\vec{v}$ and $\vec{F}$?

27.9 •• A group of particles is traveling in a magnetic field of unknown magnitude and direction. You observe that a proton moving at 1.50 km/s in the $+x$ -direction experiences a force of $2.25 \times 10^{-16} \text{ N}$ in the $+y$ -direction, and an electron moving at 4.75 km/s in the $-z$ -direction experiences a force of $8.50 \times 10^{-16} \text{ N}$ in the $+y$ -direction. (a) What are the magnitude and direction of the magnetic field? (b) What are the magnitude and direction of the magnetic force on an electron moving in the $-y$ -direction at 3.20 km/s?

Section 27.3 Magnetic Field Lines and Magnetic Flux

27.10 • A flat, square surface with side length 3.40 cm is in the xy -plane at $z = 0$. Calculate the magnitude of the flux through this surface produced by a magnetic field $\vec{B} = (0.200 \text{ T})\hat{i} + (0.300 \text{ T})\hat{j} - (0.500 \text{ T})\hat{k}$.

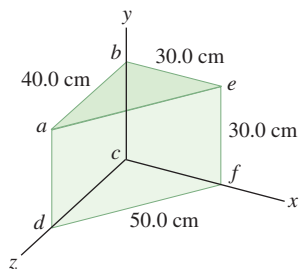
27.11 • A circular area with a radius of 6.50 cm lies in the xy -plane. What is the magnitude of the magnetic flux through this circle due to a uniform magnetic field $B = 0.230 \text{ T}$ (a) in the $+z$ -direction; (b) at an angle of 53.1° from the $+z$ -direction; (c) in the $+y$ -direction?

27.12 • A horizontal rectangular surface has dimensions 2.80 cm by 3.20 cm and is in a uniform magnetic field that is directed at an angle of 30.0° above the horizontal. What must the magnitude of the magnetic field be in order to produce a flux of $4.20 \times 10^{-4} \text{ Wb}$ through the surface?

27.13 •• An open plastic soda bottle with an opening diameter of 2.5 cm is placed on a table. A uniform 1.75-T magnetic field directed upward and oriented 25° from vertical encompasses the bottle. What is the total magnetic flux through the plastic of the soda bottle?

27.14 •• The magnetic field $\vec{B}$ in a certain region is 0.128 T, and its direction is that of the $+z$ -axis in Fig. E27.14. (a) What is the magnetic flux across the surface $abcd$ in the figure? (b) What is the magnetic flux across the surface $befc$? (c) What is the magnetic flux across the surface $aefd$? (d) What is the net flux through all five surfaces that enclose the shaded volume?

Figure E27.14

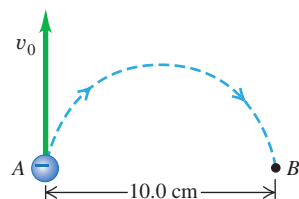


Section 27.4 Motion of Charged Particles in a Magnetic Field

27.15 •• An electron at point A in Fig. E27.15 has a speed v_0 of $1.41 \times 10^6 \text{ m/s}$. Find (a) the magnitude and direction of the magnetic field that will cause the electron to follow the semi-circular path from A to B, and (b) the time required for the electron to move from A to B.

27.16 •• Repeat Exercise 27.15 for the case in which the particle is a proton rather than an electron.

Figure E27.15



27.17 • CP A 150-g ball containing 4.00×10^8 excess electrons is dropped into a 125-m vertical shaft. At the bottom of the shaft, the ball suddenly enters a uniform horizontal magnetic field that has magnitude 0.250 T and direction from east to west. If air resistance is negligibly small, find the magnitude and direction of the force that this magnetic field exerts on the ball just as it enters the field.

27.18 • An alpha particle (a He nucleus, containing two protons and two neutrons and having a mass of $6.64 \times 10^{-27} \text{ kg}$) traveling horizontally at 35.6 km/s enters a uniform, vertical, 1.10-T magnetic field. (a) What is the diameter of the path followed by this alpha particle? (b) What effect does the magnetic field have on the speed of the particle? (c) What are the magnitude and direction of the acceleration of the alpha particle while it is in the magnetic field? (d) Explain why the speed of the particle does not change even though an unbalanced external force acts on it.

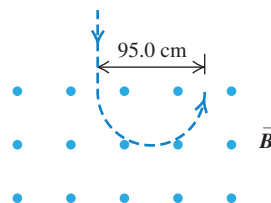
27.19 • CP A particle with charge $6.40 \times 10^{-19} \text{ C}$ travels in a circular orbit with radius 4.68 mm due to the force exerted on it by a magnetic field with magnitude 1.65 T and perpendicular to the orbit. (a) What is the magnitude of the linear momentum $\vec{p}$ of the particle? (b) What is the magnitude of the angular momentum $\vec{L}$ of the particle?

27.20 • (a) An ^{16}O nucleus (charge $+8e$) moving horizontally from west to east with a speed of 500 km/s experiences a magnetic force of 0.00320 nN vertically downward. Find the magnitude and direction of the weakest magnetic field required to produce this force. Explain how this same force could be caused by a larger magnetic field. (b) An electron moves in a uniform, horizontal, 2.10-T magnetic field that is toward the west. What must the magnitude and direction of the minimum velocity of the electron be so that the magnetic force on it will be 4.60 pN, vertically upward? Explain how the velocity could be greater than this minimum value and the force still have this same magnitude and direction.

27.21 • A deuteron (the nucleus of an isotope of hydrogen) has a mass of $3.34 \times 10^{-27} \text{ kg}$ and a charge of $+e$. The deuteron travels in a circular path with a radius of 6.96 mm in a magnetic field with magnitude 2.50 T. (a) Find the speed of the deuteron. (b) Find the time required for it to make half a revolution. (c) Through what potential difference would the deuteron have to be accelerated to acquire this speed?

27.22 • In an experiment with cosmic rays, a vertical beam of particles that have charge of magnitude $3e$ and mass 12 times the proton mass enters a uniform horizontal magnetic field of 0.250 T and is bent in a semicircle of diameter 95.0 cm, as shown in Fig. E27.22.

Figure E27.22

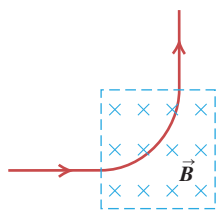


(a) Find the speed of the particles and the sign of their charge. (b) Is it reasonable to ignore the gravity force on the particles? (c) How does the speed of the particles as they enter the field compare to their speed as they exit the field?

27.23 • A physicist wishes to produce electromagnetic waves of frequency 3.0 THz (1 THz = 1 terahertz = 10^{12} Hz) using a magnetron (see Example 27.3). (a) What magnetic field would be required? Compare this field with the strongest constant magnetic fields yet produced on earth, about 45 T. (b) Would there be any advantage to using protons instead of electrons in the magnetron? Why or why not?

27.24 •• A beam of protons traveling at 1.20 km/s enters a uniform magnetic field, traveling perpendicular to the field. The beam exits the magnetic field, leaving the field in a direction perpendicular to its original direction (Fig. E27.24). The beam travels a distance of 1.18 cm while in the field. What is the magnitude of the magnetic field?

Figure E27.24



27.25 • An electron in the beam of a TV picture tube is accelerated by a potential difference of 2.00 kV. Then it passes through a region of transverse magnetic field, where it moves in a circular arc with radius 0.180 m. What is the magnitude of the field?

27.26 • A singly charged ion of ${}^7\text{Li}$ (an isotope of lithium) has a mass of 1.16×10^{-26} kg. It is accelerated through a potential difference of 220 V and then enters a magnetic field with magnitude 0.723 T perpendicular to the path of the ion. What is the radius of the ion's path in the magnetic field?

27.27 •• A proton ($q = 1.60 \times 10^{-19}$ C, $m = 1.67 \times 10^{-27}$ kg) moves in a uniform magnetic field $\vec{B} = (0.500 \text{ T})\hat{i}$. At $t = 0$ the proton has velocity components $v_x = 1.50 \times 10^5$ m/s, $v_y = 0$, and $v_z = 2.00 \times 10^5$ m/s (see Example 27.4). (a) What are the magnitude and direction of the magnetic force acting on the proton? In addition to the magnetic field there is a uniform electric field in the $+x$ -direction, $\vec{E} = (+2.00 \times 10^4 \text{ V/m})\hat{i}$. (b) Will the proton have a component of acceleration in the direction of the electric field? (c) Describe the path of the proton. Does the electric field affect the radius of the helix? Explain. (d) At $t = T/2$, where T is the period of the circular motion of the proton, what is the x -component of the displacement of the proton from its position at $t = 0$?

Section 27.5 Applications of Motion of Charged Particles

27.28 • (a) What is the speed of a beam of electrons when the simultaneous influence of an electric field of 1.56×10^4 V/m and a magnetic field of 4.62×10^{-3} T, with both fields normal to the beam and to each other, produces no deflection of the electrons? (b) In a diagram, show the relative orientation of the vectors $\vec{v}$, $\vec{E}$, and $\vec{B}$. (c) When the electric field is removed, what is the radius of the electron orbit? What is the period of the orbit?

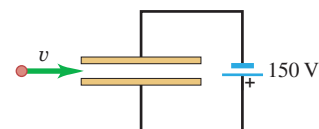
27.29 • In designing a velocity selector that uses uniform perpendicular electric and magnetic fields, you want to select positive ions of charge $+5e$ that are traveling perpendicular to the fields at 8.75 km/s. The magnetic field available to you has a magnitude of 0.550 T. (a) What magnitude of electric field do you need? (b) Show how the two fields should be oriented relative to each other and to the velocity of the ions. (c) Will your velocity selector also allow the following ions (having the same velocity as the $+5e$ ions) to pass through undeflected: (i) negative ions of charge $-5e$, (ii) positive ions of charge different from $+5e$?

27.30 • **Crossed $\vec{E}$ and $\vec{B}$ Fields.** A particle with initial velocity $\vec{v}_0 = (5.85 \times 10^3 \text{ m/s})\hat{j}$ enters a region of uniform electric and magnetic fields. The magnetic field in the region is $\vec{B} = -(1.35 \text{ T})\hat{k}$. Calculate the magnitude and direction of the electric field in the region if the particle is to pass through undeflected, for a particle of charge (a) $+0.640$ nC and (b) -0.320 nC. You can ignore the weight of the particle.

27.31 •• A 150-V battery is connected across two parallel metal plates of area 28.5 cm^2 and separation 8.20 mm. A beam of alpha particles (charge $+2e$, mass 6.64×10^{-27} kg) is accelerated from

rest through a potential difference of 1.75 kV and enters the region between the plates perpendicular to the electric field, as shown in Fig. E27.31. What magnitude and direction of magnetic field are needed so that the alpha particles emerge undeflected from between the plates?

Figure E27.31



27.32 • A singly ionized (one electron removed) ${}^{40}\text{K}$ atom passes through a velocity selector consisting of uniform perpendicular electric and magnetic fields. The selector is adjusted to allow ions having a speed of 4.50 km/s to pass through undeflected when the magnetic field is 0.0250 T. The ions next enter a second uniform magnetic field (B') oriented at right angles to their velocity. ${}^{40}\text{K}$ contains 19 protons and 21 neutrons and has a mass of 6.64×10^{-26} kg. (a) What is the magnitude of the electric field in the velocity selector? (b) What must be the magnitude of B' so that the ions will be bent into a semicircle of radius 12.5 cm?

27.33 • Singly ionized (one electron removed) atoms are accelerated and then passed through a velocity selector consisting of perpendicular electric and magnetic fields. The electric field is 155 V/m and the magnetic field is 0.0315 T. The ions next enter a uniform magnetic field of magnitude 0.0175 T that is oriented perpendicular to their velocity. (a) How fast are the ions moving when they emerge from the velocity selector? (b) If the radius of the path of the ions in the second magnetic field is 17.5 cm, what is their mass?

27.34 • In the Bainbridge mass spectrometer (see Fig. 27.24), the magnetic-field magnitude in the velocity selector is 0.650 T, and ions having a speed of 1.82×10^6 m/s pass through undeflected. (a) What is the electric-field magnitude in the velocity selector? (b) If the separation of the plates is 5.20 mm, what is the potential difference between plates P and P' ?

27.35 •• BIO Ancient Meat Eating. The amount of meat in prehistoric diets can be determined by measuring the ratio of the isotopes nitrogen-15 to nitrogen-14 in bone from human remains. Carnivores concentrate ${}^{15}\text{N}$, so this ratio tells archaeologists how much meat was consumed by ancient people. Use the spectrometer of Exercise 27.34 to find the separation of the ${}^{14}\text{N}$ and ${}^{15}\text{N}$ isotopes at the detector. The measured masses of these isotopes are 2.32×10^{-26} kg (${}^{14}\text{N}$) and 2.49×10^{-26} kg (${}^{15}\text{N}$).

Section 27.6 Magnetic Force on a Current-Carrying Conductor

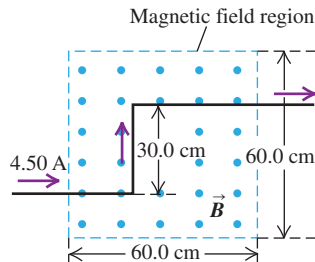
27.36 • A straight, 2.5-m wire carries a typical household current of 1.5 A (in one direction) at a location where the earth's magnetic field is 0.55 gauss from south to north. Find the magnitude and direction of the force that our planet's magnetic field exerts on this wire if it is oriented so that the current in it is running (a) from west to east, (b) vertically upward, (c) from north to south. (d) Is the magnetic force ever large enough to cause significant effects under normal household conditions?

27.37 • A straight, 2.00-m, 150-g wire carries a current in a region where the earth's magnetic field is horizontal with a magnitude of 0.55 gauss. (a) What is the minimum value of the current in this wire so that its weight is completely supported by the magnetic force due to earth's field, assuming that no other forces except gravity act on it? Does it seem likely that such a wire could support this size of current? (b) Show how the wire would have to be oriented relative to the earth's magnetic field to be supported in this way.

27.38 •• An electromagnet produces a magnetic field of 0.550 T in a cylindrical region of radius 2.50 cm between its poles. A straight wire carrying a current of 10.8 A passes through the center of this region and is perpendicular to both the axis of the cylindrical region and the magnetic field. What magnitude of force is exerted on the wire?

27.39 •• A long wire carrying 4.50 A of current makes two 90° bends, as shown in Fig. E27.39. The bent part of the wire passes through a uniform 0.240-T magnetic field directed as shown in the figure and confined to a limited region of space. Find the magnitude and direction of the force that the magnetic field exerts on the wire.

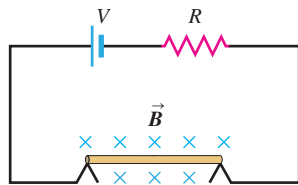
Figure E27.39



27.40 • A straight, vertical wire carries a current of 1.20 A downward in a region between the poles of a large superconducting electromagnet, where the magnetic field has magnitude $B = 0.588$ T and is horizontal. What are the magnitude and direction of the magnetic force on a 1.00-cm section of the wire that is in this uniform magnetic field, if the magnetic field direction is (a) east; (b) south; (c) 30.0° south of west?

27.41 • A thin, 50.0-cm-long metal bar with mass 750 g rests on, but is not attached to, two metallic supports in a uniform 0.450-T magnetic field, as shown in Fig. E27.41. A battery and a 25.0-Ω resistor in series are connected to the supports. (a) What is the highest voltage the battery can have without breaking the circuit at the supports? (b) The battery voltage has the maximum value calculated in part (a). If the resistor suddenly gets partially short-circuited, decreasing its resistance to 2.0 Ω, find the initial acceleration of the bar.

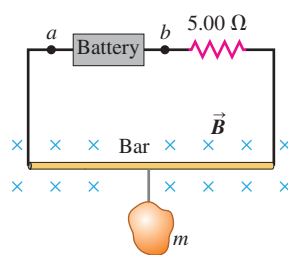
Figure E27.41



27.42 • Magnetic Balance.

The circuit shown in Fig. E27.42 is used to make a magnetic balance to weigh objects. The mass m to be measured is hung from the center of the bar that is in a uniform magnetic field of 1.50 T, directed into the plane of the figure. The battery voltage can be adjusted to vary the current in the circuit. The horizontal bar is 60.0 cm long and is made of extremely light-weight material. It is connected to the battery by thin vertical wires that can support no appreciable tension; all the weight of the suspended mass m is supported by the magnetic force on the bar. A resistor with $R = 5.00$ Ω is in series with the bar; the resistance of the rest of the circuit is much less than this. (a) Which point, a or b , should be the positive terminal of the battery? (b) If the maximum terminal voltage of the battery is 175 V, what is the greatest mass m that this instrument can measure?

Figure E27.42



27.43 • Consider the conductor and current in Example 27.8, but now let the magnetic field be parallel to the x -axis. (a) What are the magnitude and direction of the total magnetic force on the conductor? (b) In Example 27.8, the total force is the same as if we replaced the semicircle with a straight segment along the x -axis. Is

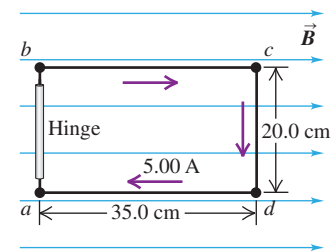
that still true when the magnetic field is in this different direction? Can you explain why, or why not?

Section 27.7 Force and Torque on a Current Loop

27.44 •• The plane of a 5.0 cm × 8.0 cm rectangular loop of wire is parallel to a 0.19-T magnetic field. The loop carries a current of 6.2 A. (a) What torque acts on the loop? (b) What is the magnetic moment of the loop? (c) What is the maximum torque that can be obtained with the same total length of wire carrying the same current in this magnetic field?

27.45 • The 20.0 cm × 35.0 cm rectangular circuit shown in Fig. E27.45 is hinged along side ab .

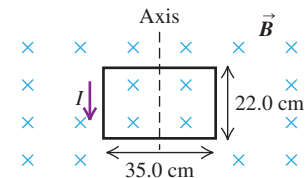
Figure E27.45



It carries a clockwise 5.00-A current and is located in a uniform 1.20-T magnetic field oriented perpendicular to two of its sides, as shown. (a) Draw a clear diagram showing the direction of the force that the magnetic field exerts on each segment of the circuit (ab , bc , etc.). (b) Of the four forces you drew in part (a), decide which ones exert a torque about the hinge ab . Then calculate only those forces that exert this torque. (c) Use your results from part (b) to calculate the torque that the magnetic field exerts on the circuit about the hinge axis ab .

27.46 • A rectangular coil of wire, 22.0 cm by 35.0 cm and carrying a current of 1.40 A, is oriented with the plane of its loop perpendicular to a uniform 1.50-T magnetic field, as shown in Fig. E27.46. (a) Calculate the net force and torque that the magnetic field exerts on the coil.

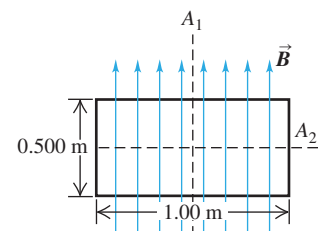
Figure E27.46



(b) The coil is rotated through a 30.0° angle about the axis shown, with the left side coming out of the plane of the figure and the right side going into the plane. Calculate the net force and torque that the magnetic field now exerts on the coil. (Hint: In order to help visualize this three-dimensional problem, make a careful drawing of the coil as viewed along the rotation axis.)

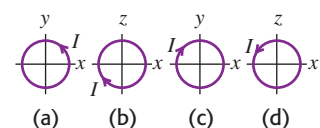
27.47 • CP A uniform rectangular coil of total mass 212 g and dimensions 0.500 m × 1.00 m is oriented with its plane parallel to a uniform 3.00-T magnetic field (Fig. E27.47). A current of 2.00 A is suddenly started in the coil. (a) About which axis (A_1 or A_2) will the coil begin to rotate? Why? (b) Find the initial angular acceleration of the coil just after the current is started.

Figure E27.47



27.48 • A circular coil with area A and N turns is free to rotate about a diameter that coincides with the x -axis. Current I is circulating in the coil. There is a uniform magnetic field $\vec{B}$ in the positive y -direction. Calculate the magnitude and direction of the torque $\vec{\tau}$

Figure E27.48



and the value of the potential energy U , as given in Eq. (27.27), when the coil is oriented as shown in parts (a) through (d) of Fig. E27.48.

27.49 •• A coil with magnetic moment $1.45 \text{ A} \cdot \text{m}^2$ is oriented initially with its magnetic moment antiparallel to a uniform 0.835-T magnetic field. What is the change in potential energy of the coil when it is rotated 180° so that its magnetic moment is parallel to the field?

Section 27.8 The Direct-Current Motor

27.50 • A dc motor with its rotor and field coils connected in series has an internal resistance of 3.2Ω . When the motor is running at full load on a 120-V line, the emf in the rotor is 105 V . (a) What is the current drawn by the motor from the line? (b) What is the power delivered to the motor? (c) What is the mechanical power developed by the motor?

27.51 •• In a shunt-wound dc motor with the field coils and rotor connected in parallel (Fig. E27.51), the resistance R_f of the field coils is 106Ω , and the resistance R_r of the rotor is 5.9Ω . When a potential difference of 120 V is applied to the brushes and the motor is running at full speed delivering mechanical power, the current supplied to it is 4.82 A . (a) What is the current in the field coils? (b) What is the current in the rotor? (c) What is the induced emf developed by the motor? (d) How much mechanical power is developed by this motor?

27.52 • A shunt-wound dc motor with the field coils and rotor connected in parallel (see Fig. E27.51) operates from a 120-V dc power line. The resistance of the field windings, R_f , is 218Ω . The resistance of the rotor, R_r , is 5.9Ω . When the motor is running, the rotor develops an emf $\mathcal{E}$. The motor draws a current of 4.82 A from the line. Friction losses amount to 45.0 W . Compute (a) the field current; (b) the rotor current; (c) the emf $\mathcal{E}$; (d) the rate of development of thermal energy in the field windings; (e) the rate of development of thermal energy in the rotor; (f) the power input to the motor; (g) the efficiency of the motor.

Section 27.9 The Hall Effect

27.53 • Figure E27.53 shows a portion of a silver ribbon with $z_1 = 11.8 \text{ mm}$ and $y_1 = 0.23 \text{ mm}$, carrying a current of 120 A in the $+x$ -direction. The ribbon lies in a uniform magnetic field, in the y -direction, with magnitude 0.95 T . Apply the simplified model of the Hall effect presented in Section 27.9. If there are 5.85×10^{28} free electrons per cubic meter, find (a) the magnitude of the drift velocity of the electrons in the x -direction; (b) the magnitude and direction of the electric field in the z -direction due to the Hall effect; (c) the Hall emf.

27.54 • Let Fig. E27.53 represent a strip of an unknown metal of the same dimensions as those of the silver ribbon in Exercise 27.53. When the magnetic field is 2.29 T and the current is 78.0 A , the Hall emf is found to be $131 \mu\text{V}$. What does the simplified model of the Hall effect presented in Section 27.9 give for the density of free electrons in the unknown metal?

Figure E27.51

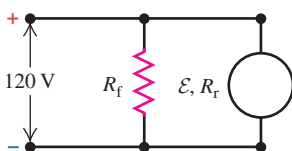
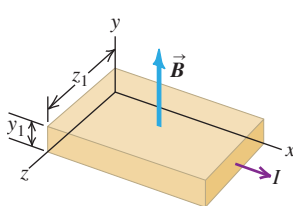


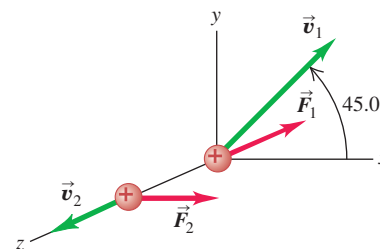
Figure E27.53



PROBLEMS

27.55 • When a particle of charge $q > 0$ moves with a velocity of $\vec{v}_1$ at 45.0° from the $+x$ -axis in the xy -plane, a uniform magnetic field exerts a force $\vec{F}_1$ along the $-z$ -axis (Fig. P27.55). When the same particle moves with a velocity $\vec{v}_2$ with the same magnitude as $\vec{v}_1$ but along the $+z$ -axis, a force $\vec{F}_2$ of magnitude F_2 is exerted on it along the $+x$ -axis. (a) What are the magnitude (in terms of q , v_1 , and F_2) and direction of the magnetic field? (b) What is the magnitude of $\vec{F}_1$ in terms of F_2 ?

Figure P27.55



27.56 • A particle with charge $9.45 \times 10^{-8} \text{ C}$ is moving in a region where there is a uniform magnetic field of 0.650 T in the $+x$ -direction. At a particular instant of time the velocity of the particle has components $v_x = -1.68 \times 10^4 \text{ m/s}$, $v_y = -3.11 \times 10^4 \text{ m/s}$, and $v_z = 5.85 \times 10^4 \text{ m/s}$. What are the components of the force on the particle at this time?

27.57 •• CP Fusion Reactor. If two deuterium nuclei (charge $+e$, mass $3.34 \times 10^{-27} \text{ kg}$) get close enough together, the attraction of the strong nuclear force will fuse them to make an isotope of helium, releasing vast amounts of energy. The range of this force is about 10^{-15} m . This is the principle behind the fusion reactor. The deuterium nuclei are moving much too fast to be contained by physical walls, so they are confined magnetically. (a) How fast would two nuclei have to move so that in a head-on collision they would get close enough to fuse? (Assume their speeds are equal. Treat the nuclei as point charges, and assume that a separation of $1.0 \times 10^{-15} \text{ m}$ is required for fusion.) (b) What strength magnetic field is needed to make deuterium nuclei with this speed travel in a circle of diameter 2.50 m ?

27.58 •• Magnetic Moment of the Hydrogen Atom. In the Bohr model of the hydrogen atom (see Section 38.5), in the lowest energy state the electron orbits the proton at a speed of $2.2 \times 10^6 \text{ m/s}$ in a circular orbit of radius $5.3 \times 10^{-11} \text{ m}$. (a) What is the orbital period of the electron? (b) If the orbiting electron is considered to be a current loop, what is the current I ? (c) What is the magnetic moment of the atom due to the motion of the electron?

27.59 •• You wish to hit a target from several meters away with a charged coin having a mass of 4.25 g and a charge of $+2500 \mu\text{C}$. The coin is given an initial velocity of 12.8 m/s , and a downward, uniform electric field with field strength 27.5 N/C exists throughout the region. If you aim directly at the target and fire the coin horizontally, what magnitude and direction of uniform magnetic field are needed in the region for the coin to hit the target?

27.60 • A cyclotron is to accelerate protons to an energy of 5.4 MeV . The superconducting electromagnet of the cyclotron produces a 2.9-T magnetic field perpendicular to the proton orbits. (a) When the protons have achieved a kinetic energy of 2.7 MeV , what is the radius of their circular orbit and what is their angular speed? (b) Repeat part (a) when the protons have achieved their final kinetic energy of 5.4 MeV .

27.61 • The magnetic poles of a small cyclotron produce a magnetic field with magnitude 0.85 T. The poles have a radius of 0.40 m, which is the maximum radius of the orbits of the accelerated particles. (a) What is the maximum energy to which protons ($q = 1.60 \times 10^{-19}$ C, $m = 1.67 \times 10^{-27}$ kg) can be accelerated by this cyclotron? Give your answer in electron volts and in joules. (b) What is the time for one revolution of a proton orbiting at this maximum radius? (c) What would the magnetic-field magnitude have to be for the maximum energy to which a proton can be accelerated to be twice that calculated in part (a)? (d) For $B = 0.85$ T, what is the maximum energy to which alpha particles ($q = 3.20 \times 10^{-19}$ C, $m = 6.65 \times 10^{-27}$ kg) can be accelerated by this cyclotron? How does this compare to the maximum energy for protons?

27.62 •• A particle with charge q is moving with speed v in the $-y$ -direction. It is moving in a uniform magnetic field $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$. (a) What are the components of the force $\vec{F}$ exerted on the particle by the magnetic field? (b) If $q > 0$, what must the signs of the components of $\vec{B}$ be if the components of $\vec{F}$ are all nonnegative? (c) If $q < 0$ and $B_x = B_y = B_z > 0$, find the direction of $\vec{F}$ and find the magnitude of $\vec{F}$ in terms of $|q|$, v , and B_x .

27.63 •• A particle with negative charge q and mass $m = 2.58 \times 10^{-15}$ kg is traveling through a region containing a uniform magnetic field $\vec{B} = -(0.120 \text{ T})\hat{k}$. At a particular instant of time the velocity of the particle is $\vec{v} = (1.05 \times 10^6 \text{ m/s})(-3\hat{i} + 4\hat{j} + 12\hat{k})$ and the force $\vec{F}$ on the particle has a magnitude of 2.45 N. (a) Determine the charge q . (b) Determine the acceleration $\vec{a}$ of the particle. (c) Explain why the path of the particle is a helix, and determine the radius of curvature R of the circular component of the helical path. (d) Determine the cyclotron frequency of the particle. (e) Although helical motion is not periodic in the full sense of the word, the x - and y -coordinates do vary in a periodic way. If the coordinates of the particle at $t = 0$ are $(x, y, z) = (R, 0, 0)$, determine its coordinates at a time $t = 2T$, where T is the period of the motion in the xy -plane.

27.64 •• **BIO Medical Uses of Cyclotrons.** The largest cyclotron in the United States is the *Tevatron* at Fermilab, near Chicago, Illinois. It is called a *Tevatron* because it can accelerate particles to energies in the TeV range: 1 tera-eV = 10^{12} eV. Its circumference is 6.4 km, and it currently can produce a maximum energy of 2.0 TeV. In a certain medical experiment, protons will be accelerated to energies of 1.25 MeV and aimed at a tumor to destroy its cells. (a) How fast are these protons moving when they hit the tumor? (b) How strong must the magnetic field be to bend the protons in the circle indicated?

27.65 • A magnetic field exerts a torque τ on a round current-carrying loop of wire. What will be the torque on this loop (in terms of τ) if its diameter is tripled?

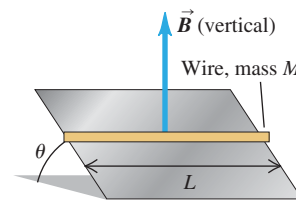
27.66 •• A particle of charge $q > 0$ is moving at speed v in the $+z$ -direction through a region of uniform magnetic field $\vec{B}$. The magnetic force on the particle is $\vec{F} = F_0(3\hat{i} + 4\hat{j})$, where F_0 is a positive constant. (a) Determine the components B_x , B_y , and B_z , or at least as many of the three components as is possible from the information given. (b) If it is given in addition that the magnetic field has magnitude $6F_0/qv$, determine as much as you can about the remaining components of $\vec{B}$.

27.67 •• Suppose the electric field between the plates in Fig. 27.24 is 1.88×10^4 V/m and the magnetic field in both regions is 0.682 T. If the source contains the three isotopes of krypton, ^{82}Kr , ^{84}Kr , and ^{86}Kr , and the ions are singly charged, find the distance between the lines formed by the three isotopes on the particle

detector. Assume the atomic masses of the isotopes (in atomic mass units) are equal to their mass numbers, 82, 84, and 86. (One atomic mass unit = $1 \text{ u} = 1.66 \times 10^{-27}$ kg.)

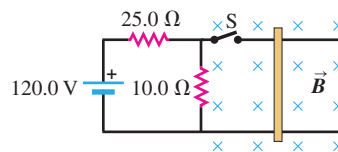
27.68 •• **Mass Spectrograph.** A mass spectrograph is used to measure the masses of ions, or to separate ions of different masses (see Section 27.5). In one design for such an instrument, ions with mass m and charge q are accelerated through a potential difference V . They then enter a uniform magnetic field that is perpendicular to their velocity, and they are deflected in a semicircular path of radius R . A detector measures where the ions complete the semicircle and from this it is easy to calculate R . (a) Derive the equation for calculating the mass of the ion from measurements of B , V , R , and q . (b) What potential difference V is needed so that singly ionized ^{12}C atoms will have $R = 50.0$ cm in a 0.150-T magnetic field? (c) Suppose the beam consists of a mixture of ^{12}C and ^{14}C ions. If v and B have the same values as in part (b), calculate the separation of these two isotopes at the detector. Do you think that this beam separation is sufficient for the two ions to be distinguished? (Make the assumption described in Problem 27.67 for the masses of the ions.)

27.69 •• A straight piece of conducting wire with mass M and length L is placed on a frictionless incline tilted at an angle θ from the horizontal (Fig. P27.69). There is a uniform, vertical magnetic field $\vec{B}$ at all points (produced by an arrangement of magnets not shown in the figure). To keep the wire from sliding down the incline, a voltage source is attached to the ends of the wire. When just the right amount of current flows through the wire, the wire remains at rest. Determine the magnitude and direction of the current in the wire that will cause the wire to remain at rest. Copy the figure and draw the direction of the current on your copy. In addition, show in a free-body diagram all the forces that act on the wire.



27.70 •• **CP** A 2.60-N metal bar, 1.50 m long and having a resistance of 10.0Ω , rests horizontally on conducting wires connecting it to the circuit shown in Fig. P27.70. The bar is in a uniform, horizontal, 1.60-T magnetic field and is not attached to the wires in the circuit. What is the acceleration of the bar just after the switch S is closed?

Figure P27.70



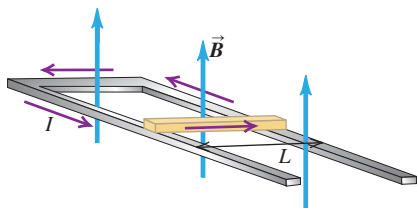
27.71 •• **Using Gauss's Law for Magnetism.** In a certain region of space, the magnetic field $\vec{B}$ is not uniform. The magnetic field has both a z -component and a component that points radially away from or toward the z -axis. The z -component is given by $B_z(z) = \beta z$, where β is a positive constant. The radial component B_r depends only on r , the radial distance from the z -axis. (a) Use Gauss's law for magnetism, Eq. (27.8), to find the radial component B_r as a function of r . (Hint: Try a cylindrical Gaussian surface of radius r concentric with the z -axis, with one end at $z = 0$ and the other at $z = L$.) (b) Sketch the magnetic field lines.

27.72 •• CP A plastic circular loop has radius R , and a positive charge q is distributed uniformly around the circumference of the loop. The loop is then rotated around its central axis, perpendicular to the plane of the loop, with angular speed ω . If the loop is in a region where there is a uniform magnetic field $\vec{B}$ directed parallel to the plane of the loop, calculate the magnitude of the magnetic torque on the loop.

27.73 •• BIO Determining Diet. One method for determining the amount of corn in early Native American diets is the *stable isotope ratio analysis* (SIRA) technique. As corn photosynthesizes, it concentrates the isotope carbon-13, whereas most other plants concentrate carbon-12. Overreliance on corn consumption can then be correlated with certain diseases, because corn lacks the essential amino acid lysine. Archaeologists use a mass spectrometer to separate the ^{12}C and ^{13}C isotopes in samples of human remains. Suppose you use a velocity selector to obtain singly ionized (missing one electron) atoms of speed 8.50 km/s , and you want to bend them within a uniform magnetic field in a semicircle of diameter 25.0 cm for the ^{12}C . The measured masses of these isotopes are $1.99 \times 10^{-26}\text{ kg}$ (^{12}C) and $2.16 \times 10^{-26}\text{ kg}$ (^{13}C). (a) What strength of magnetic field is required? (b) What is the diameter of the ^{13}C semicircle? (c) What is the separation of the ^{12}C and ^{13}C ions at the detector at the end of the semicircle? Is this distance large enough to be easily observed?

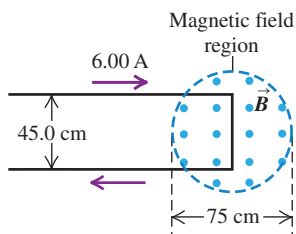
27.74 •• CP An Electromagnetic Rail Gun. A conducting bar with mass m and length L slides over horizontal rails that are connected to a voltage source. The voltage source maintains a constant current I in the rails and bar, and a constant, uniform, vertical magnetic field $\vec{B}$ fills the region between the rails (Fig. P27.74). (a) Find the magnitude and direction of the net force on the conducting bar. Ignore friction, air resistance, and electrical resistance. (b) If the bar has mass m , find the distance d that the bar must move along the rails from rest to attain speed v . (c) It has been suggested that rail guns based on this principle could accelerate payloads into earth orbit or beyond. Find the distance the bar must travel along the rails if it is to reach the escape speed for the earth (11.2 km/s). Let $B = 0.80\text{ T}$, $I = 2.0 \times 10^3\text{ A}$, $m = 25\text{ kg}$, and $L = 50\text{ cm}$. For simplicity assume the net force on the object is equal to the magnetic force, as in parts (a) and (b), even though gravity plays an important role in an actual launch in space.

Figure P27.74



27.75 • A long wire carrying a 6.00-A current reverses direction by means of two right-angle bends, as shown in Fig. P27.75. The part of the wire where the bend occurs is in a magnetic field of 0.666 T confined to the circular region of diameter 75 cm , as shown. Find the magnitude and direction of the net force that the magnetic field exerts on this wire.

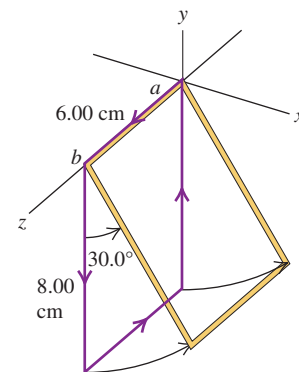
Figure P27.75



27.76 • A wire 25.0 cm long lies along the z -axis and carries a current of 7.40 A in the $+z$ -direction. The magnetic field is uniform and has components $B_x = -0.242\text{ T}$, $B_y = -0.985\text{ T}$, and $B_z = -0.336\text{ T}$. (a) Find the components of the magnetic force on the wire. (b) What is the magnitude of the net magnetic force on the wire?

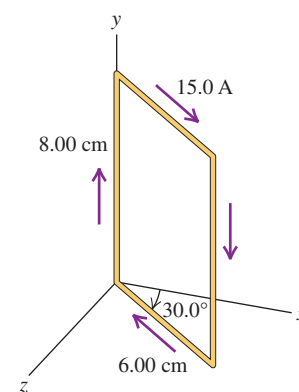
27.77 •• CP The rectangular loop of wire shown in Fig. P27.77 has a mass of 0.15 g per centimeter of length and is pivoted about side ab on a frictionless axis. The current in the wire is 8.2 A in the direction shown. Find the magnitude and direction of the magnetic field parallel to the y -axis that will cause the loop to swing up until its plane makes an angle of 30.0° with the yz -plane.

Figure P27.77



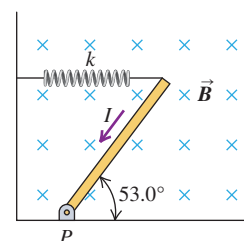
27.78 •• The rectangular loop shown in Fig. P27.78 is pivoted about the y -axis and carries a current of 15.0 A in the direction indicated. (a) If the loop is in a uniform magnetic field with magnitude 0.48 T in the $+x$ -direction, find the magnitude and direction of the torque required to hold the loop in the position shown. (b) Repeat part (a) for the case in which the field is in the $-z$ -direction. (c) For each of the above magnetic fields, what torque would be required if the loop were pivoted about an axis through its center, parallel to the y -axis?

Figure P27.78



27.79 •• CP CALC A thin, uniform rod with negligible mass and length 0.200 m is attached to the floor by a frictionless hinge at point P (Fig. P27.79). A horizontal spring with force constant $k = 4.80\text{ N/m}$ connects the other end of the rod to a vertical wall. The rod is in a uniform magnetic field $B = 0.340\text{ T}$ directed into the plane of the figure. There is current $I = 6.50\text{ A}$ in the rod, in the direction shown. (a) Calculate the torque due to the magnetic force on the rod, for an axis at P . Is it correct to take the total magnetic force to act at the center of gravity of the rod when calculating the torque? Explain. (b) When the rod is in equilibrium and makes an angle of 53.0° with the floor, is the spring stretched or compressed? (c) How much energy is stored in the spring when the rod is in equilibrium?

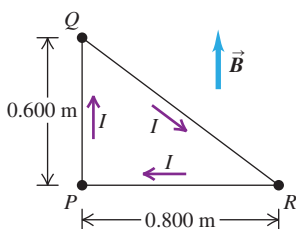
Figure P27.79



27.80 •• The loop of wire shown in Fig. P27.80 forms a right triangle and carries a current $I = 5.00\text{ A}$ in the direction shown. The loop is in a uniform magnetic field that has magnitude $B = 3.00\text{ T}$ and the same direction as the current in side PQ of the loop. (a) Find the force exerted by the magnetic field on each side of the triangle. If the force is not zero, specify its direction. (b) What is the net force on the loop? (c) The loop is pivoted about an axis that lies

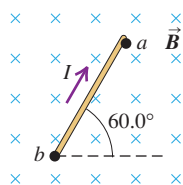
along side PR . Use the forces calculated in part (a) to calculate the torque on each side of the loop (see Problem 27.79). (d) What is the magnitude of the net torque on the loop? Calculate the net torque from the torques calculated in part (c) and also from Eq. (27.28). Do these two results agree? (e) Is the net torque directed to rotate point Q into the plane of the figure or out of the plane of the figure?

Figure P27.80



27.81 •• CP A uniform, 458-g metal bar 75.0 cm long carries a current I in a uniform, horizontal 1.25-T magnetic field as shown in Fig. P27.81. The directions of I and $\vec{B}$ are shown in the figure. The bar is free to rotate about a frictionless hinge at point b . The other end of the bar rests on a conducting support at point a but is not attached there. The bar rests at an angle of 60.0° above the horizontal. What is the largest value the current I can have without breaking the electrical contact at a ? (See Problem 27.77.)

Figure P27.81



27.82 •• Paleoclimate. Climatologists can determine the past temperature of the earth by comparing the ratio of the isotope oxygen-18 to the isotope oxygen-16 in air trapped in ancient ice sheets, such as those in Greenland. In one method for separating these isotopes, a sample containing both of them is first singly ionized (one electron is removed) and then accelerated from rest through a potential difference V . This beam then enters a magnetic field B at right angles to the field and is bent into a quarter-circle. A particle detector at the end of the path measures the amount of each isotope. (a) Show that the separation Δr of the two isotopes at the detector is given by

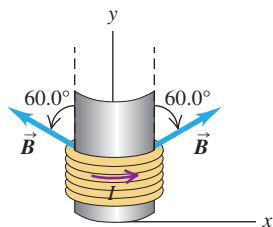
$$\Delta r = \frac{\sqrt{2eV}}{eB} (\sqrt{m_{18}} - \sqrt{m_{16}})$$

where m_{16} and m_{18} are the masses of the two oxygen isotopes, (b) The measured masses of the two isotopes are 2.66×10^{-26} kg (^{16}O) and 2.99×10^{-26} kg (^{18}O). If the magnetic field is 0.050 T, what must be the accelerating potential V so that these two isotopes will be separated by 4.00 cm at the detector?

27.83 •• CALC A Voice Coil. It was shown in Section 27.7 that the net force on a current loop in a uniform magnetic field is zero. The magnetic force on the voice coil of a loudspeaker (see Fig. 27.28) is nonzero because the magnetic field at the coil is not uniform. A voice coil in a loudspeaker has 50 turns of wire and a diameter of 1.56 cm, and the current in the coil is 0.950 A.

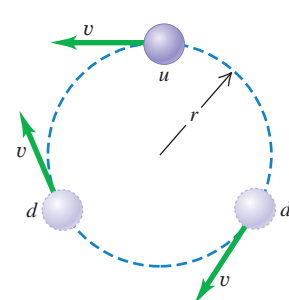
Assume that the magnetic field at each point of the coil has a constant magnitude of 0.220 T and is directed at an angle of 60.0° outward from the normal to the plane of the coil (Fig. P27.83). Let the axis of the coil be in the y -direction. The current in the coil is in the direction shown (counterclockwise as viewed from a point above the coil on the y -axis). Calculate the magnitude and direction of the net magnetic force on the coil.

Figure P27.83



27.84 •• Quark Model of the Neutron. The neutron is a particle with zero charge. Nonetheless, it has a nonzero magnetic moment with z -component 9.66×10^{-27} A·m². This can be explained by the internal structure of the neutron. A substantial body of evidence indicates that a neutron is composed of three fundamental particles called *quarks*: an “up” (u) quark, of charge $+2e/3$, and two “down” (d) quarks, each of charge $-e/3$. The combination of the three quarks produces a net charge of $2e/3 - e/3 - e/3 = 0$. If the quarks are in motion, they can produce a nonzero magnetic moment. As a very simple model, suppose the u quark moves in a counterclockwise circular path and the d quarks move in a clockwise circular path, all of radius r and all with the same speed v (Fig. P27.84).

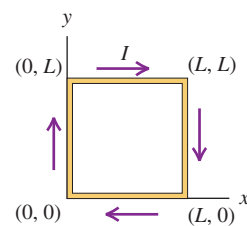
Figure P27.84



(a) Determine the current due to the circulation of the u quark. (b) Determine the magnitude of the magnetic moment due to the circulating u quark. (c) Determine the magnitude of the magnetic moment of the three-quark system. (Be careful to use the correct magnetic moment directions.) (d) With what speed v must the quarks move if this model is to reproduce the magnetic moment of the neutron? Use $r = 1.20 \times 10^{-15}$ m (the radius of the neutron) for the radius of the orbits.

27.85 •• CALC Force on a Current Loop in a Nonuniform Magnetic Field. It was shown in Section 27.7 that the net force on a current loop in a uniform magnetic field is zero. But what if $\vec{B}$ is not uniform? Figure P27.85 shows a square loop of wire that lies in the xy -plane. The loop has corners at $(0, 0)$, $(0, L)$, $(L, 0)$, and (L, L) and carries a constant current I in the clockwise direction. The magnetic field has no x -component but has both y - and z -components: $\vec{B} = (B_0 z/L)\hat{j} + (B_0 y/L)\hat{k}$, where B_0 is a positive constant. (a) Sketch the magnetic field lines in the yz -plane. (b) Find the magnitude and direction of the magnetic force exerted on each of the sides of the loop by integrating Eq. (27.20). (c) Find the magnitude and direction of the net magnetic force on the loop.

Figure P27.85

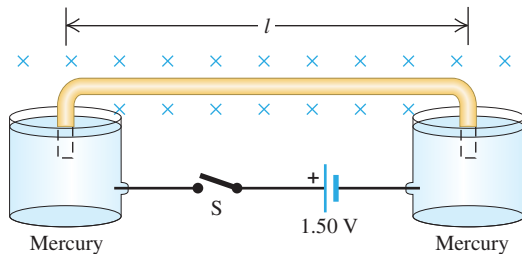


27.86 • CALC Torque on a Current Loop in a Nonuniform Magnetic Field. In Section 27.7 the expression for the torque on a current loop was derived assuming that the magnetic field $\vec{B}$ was uniform. But what if $\vec{B}$ is not uniform? Figure P27.85 shows a square loop of wire that lies in the xy -plane. The loop has corners at $(0, 0)$, $(0, L)$, $(L, 0)$, and (L, L) and carries a constant current I in the clockwise direction. The magnetic field has no z -component but has both x - and y -components: $\vec{B} = (B_0 y/L)\hat{i} + (B_0 x/L)\hat{j}$, where B_0 is a positive constant. (a) Sketch the magnetic field lines in the xy -plane. (b) Find the magnitude and direction of the magnetic force exerted on each of the sides of the loop by integrating Eq. (27.20). (c) If the loop is free to rotate about the x -axis, find the magnitude and direction of the magnetic torque on the loop. (d) Repeat part (c) for the case in which the loop is free to rotate about the y -axis. (e) Is Eq. (27.26), $\vec{\tau} = \vec{\mu} \times \vec{B}$, an appropriate description of the torque on this loop? Why or why not?

27.87 •• CP An insulated wire with mass $m = 5.40 \times 10^{-5}$ kg is bent into the shape of an inverted U such that the horizontal part has a length $l = 15.0$ cm. The bent ends of the wire are partially

immersed in two pools of mercury, with 2.5 cm of each end below the mercury's surface. The entire structure is in a region containing a uniform 0.00650-T magnetic field directed into the page (Fig. P27.87). An electrical connection from the mercury pools is made through the ends of the wires. The mercury pools are connected to a 1.50-V battery and a switch S . When switch S is closed, the wire jumps 35.0 cm into the air, measured from its initial position. (a) Determine the speed v of the wire as it leaves the mercury. (b) Assuming that the current I through the wire was constant from the time the switch was closed until the wire left the mercury, determine I . (c) Ignoring the resistance of the mercury and the circuit wires, determine the resistance of the moving wire.

Figure P27.87

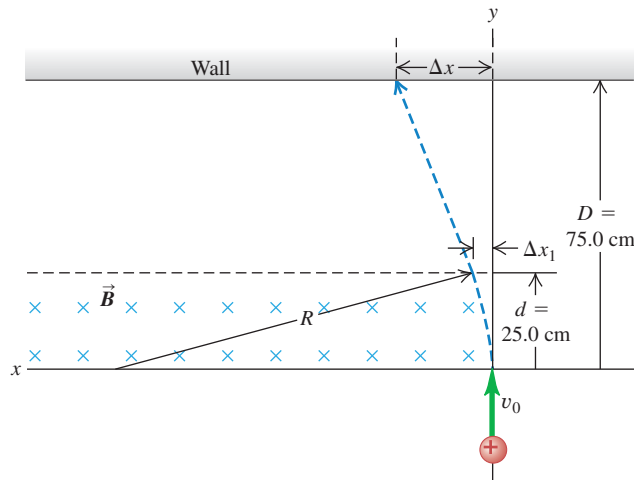


27.88 •• A circular loop of wire with area A lies in the xy -plane. As viewed along the z -axis looking in the $-z$ -direction toward the origin, a current I is circulating clockwise around the loop. The torque produced by an external magnetic field $\vec{B}$ is given by $\vec{\tau} = D(4\hat{i} - 3\hat{j})$, where D is a positive constant, and for this orientation of the loop the magnetic potential energy $U = -\vec{\mu} \cdot \vec{B}$ is negative. The magnitude of the magnetic field is $B_0 = 13D/IA$. (a) Determine the vector magnetic moment of the current loop. (b) Determine the components B_x , B_y , and B_z of $\vec{B}$.

CHALLENGE PROBLEMS

27.89 ••• A particle with charge $2.15 \mu\text{C}$ and mass $3.20 \times 10^{-11} \text{ kg}$ is initially traveling in the $+y$ -direction with a speed $v_0 = 1.45 \times 10^5 \text{ m/s}$. It then enters a region containing a uniform magnetic field that is directed into, and perpendicular to, the page in Fig. P27.89. The magnitude of the field is 0.420 T. The

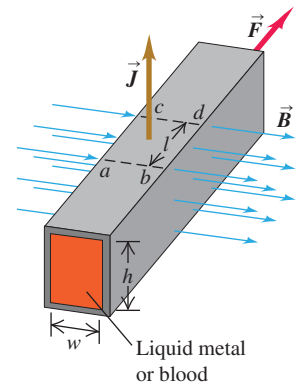
Figure P27.89



region extends a distance of 25.0 cm along the initial direction of travel; 75.0 cm from the point of entry into the magnetic field region is a wall. The length of the field-free region is thus 50.0 cm. When the charged particle enters the magnetic field, it follows a curved path whose radius of curvature is R . It then leaves the magnetic field after a time t_1 , having been deflected a distance Δx_1 . The particle then travels in the field-free region and strikes the wall after undergoing a total deflection Δx . (a) Determine the radius R of the curved part of the path. (b) Determine t_1 , the time the particle spends in the magnetic field. (c) Determine Δx_1 , the horizontal deflection at the point of exit from the field. (d) Determine Δx , the total horizontal deflection.

27.90 ••• The Electromagnetic Pump. Magnetic forces acting on conducting fluids provide a convenient means of pumping these fluids. For example, this method can be used to pump blood without the damage to the cells that can be caused by a mechanical pump. A horizontal tube with rectangular cross section (height h , width w) is placed at right angles to a uniform magnetic field with magnitude B so that a length l is in the field (Fig. P27.90). The tube is filled with a conducting liquid, and an electric current of density J is maintained in the third mutually perpendicular direction.

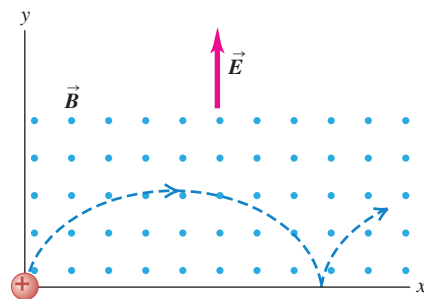
Figure P27.90



(a) Show that the difference of pressure between a point in the liquid on a vertical plane through ab and a point in the liquid on another vertical plane through cd , under conditions in which the liquid is prevented from flowing, is $\Delta p = JIB$. (b) What current density is needed to provide a pressure difference of 1.00 atm between these two points if $B = 2.20 \text{ T}$ and $l = 35.0 \text{ mm}$?

27.91 ••• CP A Cycloidal Path. A particle with mass m and positive charge q starts from rest at the origin shown in Fig. P27.91. There is a uniform electric field $\vec{E}$ in the $+y$ -direction and a uniform magnetic field $\vec{B}$ directed out of the page. It is shown in more advanced books that the path is a *cycloid* whose radius of curvature at the top points is twice the y -coordinate at that level. (a) Explain why the path has this general shape and why it is repetitive. (b) Prove that the speed at any point is equal to $\sqrt{2qEy/m}$. (Hint: Use energy conservation.) (c) Applying Newton's second law at the top point and taking as given that the radius of curvature here equals $2y$, prove that the speed at this point is $2E/B$.

Figure P27.91



Problems

For instructor-assigned homework, go to www.masteringphysics.com

•, ••, •••: Problems of increasing difficulty. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q28.1 A topic of current interest in physics research is the search (thus far unsuccessful) for an isolated magnetic pole, or magnetic *monopole*. If such an entity were found, how could it be recognized? What would its properties be?

Q28.2 Streams of charged particles emitted from the sun during periods of solar activity create a disturbance in the earth's magnetic field. How does this happen?

Q28.3 The text discussed the magnetic field of an infinitely long, straight conductor carrying a current. Of course, there is no such thing as an infinitely long *anything*. How do you decide whether a particular wire is long enough to be considered infinite?

Q28.4 Two parallel conductors carrying current in the same direction attract each other. If they are permitted to move toward each other, the forces of attraction do work. From where does the energy come? Does this contradict the assertion in Chapter 27 that magnetic forces on moving charges do no work? Explain.

Q28.5 Pairs of conductors carrying current into or out of the power-supply components of electronic equipment are sometimes twisted together to reduce magnetic-field effects. Why does this help?

Q28.6 Suppose you have three long, parallel wires arranged so that in cross section they are at the corners of an equilateral triangle. Is there any way to arrange the currents so that all three wires attract each other? So that all three wires repel each other? Explain.

Q28.7 In deriving the force on one of the long, current-carrying conductors in Section 28.4, why did we use the magnetic field due to only one of the conductors? That is, why didn't we use the *total* magnetic field due to *both* conductors?

Q28.8 Two concentric, coplanar, circular loops of wire of different diameter carry currents in the same direction. Describe the nature of the force exerted on the inner loop by the outer loop and on the outer loop by the inner loop.

Q28.9 A current was sent through a helical coil spring. The spring contracted, as though it had been compressed. Why?

Q28.10 What are the relative advantages and disadvantages of Ampere's law and the law of Biot and Savart for practical calculations of magnetic fields?

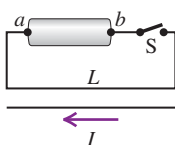
Q28.11 Magnetic field lines never have a beginning or an end. Use this to explain why it is reasonable for the field of a toroidal solenoid to be confined entirely to its interior, while a straight solenoid *must* have some field outside.

Q28.12 If the magnitude of the magnetic field a distance R from a very long, straight, current-carrying wire is B , at what distance from the wire will the field have magnitude $3B$?

Q28.13 Two very long, parallel wires carry equal currents in opposite directions. (a) Is there any place that their magnetic fields completely cancel? If so, where? If not, why not? (b) How would the answer to part (a) change if the currents were in the same direction?

Q28.14 In the circuit shown in Fig. Q28.14, when switch S is suddenly closed, the wire L is pulled toward the lower wire carrying current I . Which (a or b) is the positive terminal of the battery? How do you know?

Figure Q28.14



Q28.15 A metal ring carries a current that causes a magnetic field B_0 at the center of the ring and a field B at point P a distance x from the center along the axis of the ring. If the radius of the ring is doubled, find the magnetic field at the center. Will the field at point P change by the same factor? Why?

Q28.16 Why should the permeability of a paramagnetic material be expected to decrease with increasing temperature?

Q28.17 If a magnet is suspended over a container of liquid air, it attracts droplets to its poles. The droplets contain only liquid oxygen; even though nitrogen is the primary constituent of air, it is not attracted to the magnet. Explain what this tells you about the magnetic susceptibilities of oxygen and nitrogen, and explain why a magnet in ordinary, room-temperature air doesn't attract molecules of oxygen *gas* to its poles.

Q28.18 What features of atomic structure determine whether an element is diamagnetic or paramagnetic? Explain.

Q28.19 The magnetic susceptibility of paramagnetic materials is quite strongly temperature dependent, but that of diamagnetic materials is nearly independent of temperature. Why the difference?

Q28.20 A cylinder of iron is placed so that it is free to rotate around its axis. Initially the cylinder is at rest, and a magnetic field is applied to the cylinder so that it is magnetized in a direction parallel to its axis. If the direction of the *external* field is suddenly reversed, the direction of magnetization will also reverse and the cylinder will begin rotating around its axis. (This is called the *Einstein-de Haas effect*.) Explain why the cylinder begins to rotate.

Q28.21 The discussion of magnetic forces on current loops in Section 27.7 commented that no net force is exerted on a complete loop in a uniform magnetic field, only a torque. Yet magnetized materials that contain atomic current loops certainly *do* experience net forces in magnetic fields. How is this discrepancy resolved?

Q28.22 Show that the units $A \cdot m^2$ and J/T for the Bohr magneton are equivalent.

EXERCISES

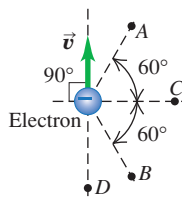
Section 26.1 Magnetic Field of a Moving Charge

28.1 •• A $+6.00\text{-}\mu\text{C}$ point charge is moving at a constant $8.00 \times 10^6 \text{ m/s}$ in the $+y$ -direction, relative to a reference frame. At the instant when the point charge is at the origin of this reference frame, what is the magnetic-field vector $\vec{B}$ it produces at the following points: (a) $x = 0.500 \text{ m}$, $y = 0$, $z = 0$; (b) $x = 0$, $y = -0.500 \text{ m}$, $z = 0$; (c) $x = 0$, $y = 0$, $z = +0.500 \text{ m}$; (d) $x = 0$, $y = -0.500 \text{ m}$, $z = +0.500 \text{ m}$?

28.2 • **Fields Within the Atom.** In the Bohr model of the hydrogen atom, the electron moves in a circular orbit of radius $5.3 \times 10^{-11} \text{ m}$ with a speed of $2.2 \times 10^6 \text{ m/s}$. If we are viewing the atom in such a way that the electron's orbit is in the plane of the paper with the electron moving clockwise, find the magnitude and direction of the electric and magnetic fields that the electron produces at the location of the nucleus (treated as a point).

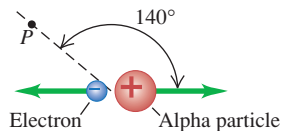
28.3 • An electron moves at $0.100c$ as shown in Fig. E28.3. Find the magnitude and direction of the magnetic field this electron produces at the following points, each $2.00\ \mu\text{m}$ from the electron: (a) points A and B; (b) point C; (c) point D.

Figure E28.3



28.4 • An alpha particle (charge $+2e$) and an electron move in opposite directions from the same point, each with the speed of $2.50 \times 10^5\ \text{m/s}$ (Fig. E28.4). Find the magnitude and direction of the total magnetic field these charges produce at point P, which is $1.75\ \text{nm}$ from each of them.

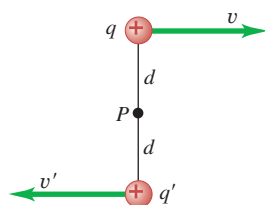
Figure E28.4



28.5 • A $-4.80\text{-}\mu\text{C}$ charge is moving at a constant speed of $6.80 \times 10^5\ \text{m/s}$ in the $+x$ -direction relative to a reference frame. At the instant when the point charge is at the origin, what is the magnetic-field vector it produces at the following points: (a) $x = 0.500\ \text{m}$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 0.500\ \text{m}$, $z = 0$; (c) $x = 0.500\ \text{m}$, $y = 0.500\ \text{m}$, $z = 0$; (d) $x = 0$, $y = 0$, $z = 0.500\ \text{m}$?

28.6 • Positive point charges $q = +8.00\ \mu\text{C}$ and $q' = +3.00\ \mu\text{C}$ are moving relative to an observer at point P, as shown in Fig. E28.6. The distance d is $0.120\ \text{m}$, $v = 4.50 \times 10^6\ \text{m/s}$, and $v' = 9.00 \times 10^6\ \text{m/s}$. (a) When the two charges are at the locations shown in the figure, what are the magnitude and direction of the net magnetic field they produce at point P? (b) What are the magnitude and direction of the electric and magnetic forces that each charge exerts on the other, and what is the ratio of the magnitude of the electric force to the magnitude of the magnetic force? (c) If the direction of $\vec{v}'$ is reversed, so both charges are moving in the same direction, what are the magnitude and direction of the magnetic forces that the two charges exert on each other?

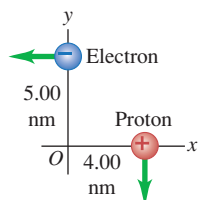
Figure E28.6



28.7 • Figure E28.6 shows two point charges, q and q' , moving relative to an observer at point P. Suppose that the lower charge is actually negative, with $q' = -q$. (a) Find the magnetic field (magnitude and direction) produced by the two charges at point P if (i) $v' = v/2$; (ii) $v' = v$; (iii) $v' = 2v$. (b) Find the direction of the magnetic force that q exerts on q' , and find the direction of the magnetic force that q' exerts on q . (c) If $v = v' = 3.00 \times 10^5\ \text{m/s}$, what is the ratio of the magnitude of the magnetic force acting on each charge to that of the Coulomb force acting on each charge?

28.8 • An electron and a proton are each moving at $845\ \text{km/s}$ in perpendicular paths as shown in Fig. E28.8. At the instant when they are at the positions shown in the figure, find the magnitude and direction of (a) the total magnetic field they produce at the origin; (b) the magnetic field the electron produces at the location of the proton; (c) the total electric force and the total magnetic force that the electron exerts on the proton.

Figure E28.8



28.9 • A negative charge $q = -3.60 \times 10^{-6}\ \text{C}$ is located at the origin and has velocity $\vec{v} = (7.50 \times 10^4\ \text{m/s})\hat{i} + (-4.90 \times 10^4\ \text{m/s})\hat{j}$. At this instant what are the magnitude and direction of

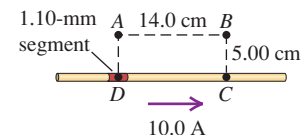
the magnetic field produced by this charge at the point $x = 0.200\ \text{m}$, $y = -0.300\ \text{m}$, $z = 0$?

Section 28.2 Magnetic Field of a Current Element

28.10 • A short current element $d\vec{l} = (0.500\ \text{mm})\hat{j}$ carries a current of $8.20\ \text{A}$ in the same direction as $d\vec{l}$. Point P is located at $\vec{r} = (-0.730\ \text{m})\hat{i} + (0.390\ \text{m})\hat{k}$. Use unit vectors to express the magnetic field at P produced by this current element.

28.11 • A straight wire carries a 10.0-A current (Fig. E28.11).

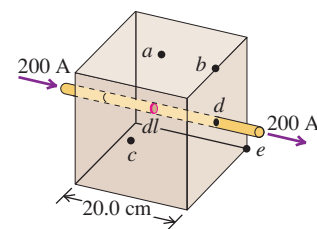
Figure E28.11



ABCD is a rectangle with point D in the middle of a 1.10-mm segment of the wire and point C in the wire. Find the magnitude and direction of the magnetic field due to this segment at (a) point A; (b) point B; (c) point C.

28.12 • A long, straight wire, carrying a current of $200\ \text{A}$, runs through a cubical wooden box, entering and leaving through holes in the centers of opposite faces (Fig. E28.12). The length of each side of the box is $20.0\ \text{cm}$. Consider an element dl of the wire $0.100\ \text{cm}$ long at the center of the box.

Figure E28.12

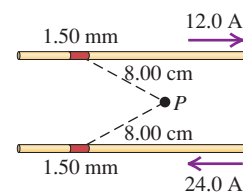


Compute the magnitude dB of the magnetic field produced by this element at the points a , b , c , d , and e in Fig. E28.12. Points a , c , and d are at the centers of the faces of the cube; point b is at the midpoint of one edge; and point e is at a corner. Copy the figure and show the directions and relative magnitudes of the field vectors. (Note: Assume that the length dl is small in comparison to the distances from the current element to the points where the magnetic field is to be calculated.)

28.13 • A long, straight wire lies along the z -axis and carries a 4.00-A current in the $+z$ -direction. Find the magnetic field (magnitude and direction) produced at the following points by a 0.500-mm segment of the wire centered at the origin: (a) $x = 2.00\ \text{m}$, $y = 0$, $z = 0$; (b) $x = 0$, $y = 2.00\ \text{m}$, $z = 0$; (c) $x = 2.00\ \text{m}$, $y = 2.00\ \text{m}$, $z = 0$; (d) $x = 0$, $y = 0$, $z = 2.00\ \text{m}$.

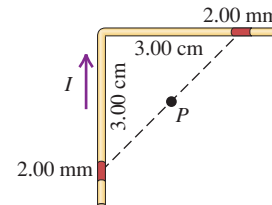
28.14 • Two parallel wires are $5.00\ \text{cm}$ apart and carry currents in opposite directions, as shown in Fig. E28.14. Find the magnitude and direction of the magnetic field at point P due to two 1.50-mm segments of wire that are opposite each other and each $8.00\ \text{cm}$ from P.

Figure E28.14



28.15 • A wire carrying a 28.0-A current bends through a right angle. Consider two 2.00-mm segments of wire, each $3.00\ \text{cm}$ from the bend (Fig. E28.15). Find the magnitude and direction of the magnetic field these two segments produce at point P, which is midway between them.

Figure E28.15



28.16 • A square wire loop $10.0\ \text{cm}$ on each side carries a clockwise current of $15.0\ \text{A}$. Find the magnitude and direction of the magnetic field at its center due to the four 1.20-mm wire segments at the midpoint of each side.

Section 28.3 Magnetic Field of a Straight Current-Carrying Conductor

28.17 • The Magnetic Field from a Lightning Bolt. Lightning bolts can carry currents up to approximately 20 kA. We can model such a current as the equivalent of a very long, straight wire. (a) If you were unfortunate enough to be 5.0 m away from such a lightning bolt, how large a magnetic field would you experience? (b) How does this field compare to one you would experience by being 5.0 cm from a long, straight household current of 10 A?

28.18 • A very long, straight horizontal wire carries a current such that 3.50×10^{18} electrons per second pass any given point going from west to east. What are the magnitude and direction of the magnetic field this wire produces at a point 4.00 cm directly above it?

28.19 • BIO Currents in the Heart. The body contains many small currents caused by the motion of ions in the organs and cells. Measurements of the magnetic field around the chest due to currents in the heart give values of about $10 \mu\text{G}$. Although the actual currents are rather complicated, we can gain a rough understanding of their magnitude if we model them as a long, straight wire. If the surface of the chest is 5.0 cm from this current, how large is the current in the heart?

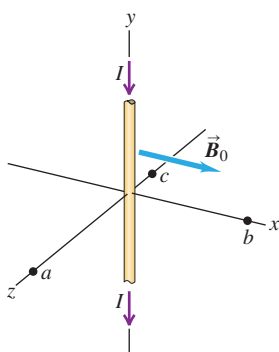
28.20 • BIO Bacteria Navigation. Certain bacteria (such as *Aquaspirillum magnetotacticum*) tend to swim toward the earth's geographic north pole because they contain tiny particles, called magnetosomes, that are sensitive to a magnetic field. If a transmission line carrying 100 A is laid underwater, at what range of distances would the magnetic field from this line be great enough to interfere with the migration of these bacteria? (Assume that a field less than 5 percent of the earth's field would have little effect on the bacteria. Take the earth's field to be $5.0 \times 10^{-5} \text{ T}$ and ignore the effects of the seawater.)

28.21 • (a) How large a current would a very long, straight wire have to carry so that the magnetic field 2.00 cm from the wire is equal to 1.00 G (comparable to the earth's northward-pointing magnetic field)? (b) If the wire is horizontal with the current running from east to west, at what locations would the magnetic field of the wire point in the same direction as the horizontal component of the earth's magnetic field? (c) Repeat part (b) except the wire is vertical with the current going upward.

28.22 • Two long, straight wires, one above the other, are separated by a distance $2a$ and are parallel to the x -axis. Let the $+y$ -axis be in the plane of the wires in the direction from the lower wire to the upper wire. Each wire carries current I in the $+x$ -direction. What are the magnitude and direction of the net magnetic field of the two wires at a point in the plane of the wires (a) midway between them; (b) at a distance a above the upper wire; (c) at a distance a below the lower wire?

28.23 •• A long, straight wire lies along the y -axis and carries a current $I = 8.00 \text{ A}$ in the $-y$ -direction (Fig. E28.23). In addition to the magnetic field due to the current in the wire, a uniform magnetic field $\vec{B}_0$ with magnitude $1.50 \times 10^{-6} \text{ T}$ is in the $+x$ -direction. What is the total field (magnitude and direction) at the following points in the xz -plane: (a) $x = 0$, $z = 1.00 \text{ m}$; (b) $x = 1.00 \text{ m}$, $z = 0$; (c) $x = 0$, $z = -0.25 \text{ m}$?

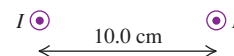
Figure E28.23



28.24 •• BIO EMF. Currents in dc transmission lines can be 100 A or more. Some people have expressed concern that the electromagnetic fields (EMFs) from such lines near their homes could cause health dangers. For a line with current 150 A and at a height of 8.0 m above the ground, what magnetic field does the line produce at ground level? Express your answer in teslas and as a percent of the earth's magnetic field, which is 0.50 gauss. Does this seem to be cause for worry?

28.25 • Two long, straight, parallel wires, 10.0 cm apart, carry equal 4.00-A currents in the same direction, as shown in Fig. E28.25. Find the magnitude and direction of the magnetic field at (a) point P_1 , midway between the wires; (b) point P_2 , 25.0 cm to the right of P_1 ; (c) point P_3 , 20.0 cm directly above P_1 .

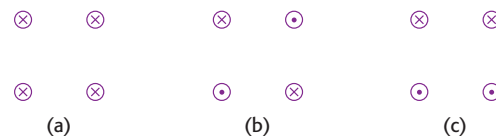
Figure E28.25



28.26 •• A rectangular loop with dimensions 4.20 cm by 9.50 cm carries current I . The current in the loop produces a magnetic field at the center of the loop that has magnitude $5.50 \times 10^{-5} \text{ T}$ and direction away from you as you view the plane of the loop. What are the magnitude and direction (clockwise or counterclockwise) of the current in the loop?

28.27 • Four, long, parallel power lines each carry 100-A currents. A cross-sectional diagram of these lines is a square, 20.0 cm on each side. For each of the three cases shown in Fig. E28.27, calculate the magnetic field at the center of the square.

Figure E28.27



28.28 • Four very long, current-carrying wires in the same plane intersect to form a square 40.0 cm on each side, as shown in Fig. E28.28. Find the magnitude and direction of the current I so that the magnetic field at the center of the square is zero.

Figure E28.28

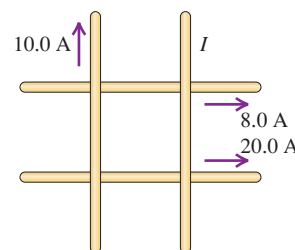
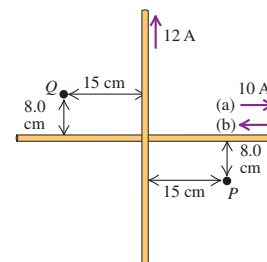


Figure E28.29

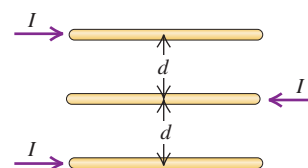


28.29 •• Two insulated wires perpendicular to each other in the same plane carry currents as shown in Fig. E28.29. Find the magnitude of the net magnetic field these wires produce at points P and Q if the 10.0 A-current is (a) to the right or (b) to the left.

Section 28.4 Force Between Parallel Conductors

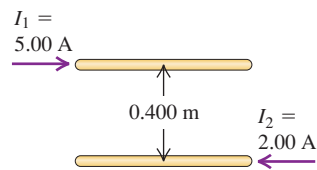
28.30 • Three parallel wires each carry current I in the directions shown in Fig. E28.30. If the separation between adjacent wires is d , calculate the magnitude and direction of the net magnetic force per unit length on each wire.

Figure E28.30



28.31 • Two long, parallel wires are separated by a distance of 0.400 m (Fig. E28.31). The currents I_1 and I_2 have the directions shown. (a) Calculate the magnitude of the force exerted by each wire on a 1.20-m length of the other. Is the force attractive or repulsive? (b) Each current is doubled, so that I_1 becomes 10.0 A and I_2 becomes 4.00 A. Now what is the magnitude of the force that each wire exerts on a 1.20-m length of the other?

Figure E28.31

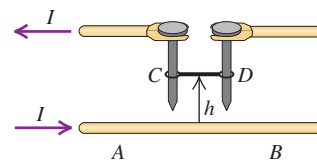


28.32 • Two long, parallel wires are separated by a distance of 2.50 cm. The force per unit length that each wire exerts on the other is 4.00×10^{-5} N/m, and the wires repel each other. The current in one wire is 0.600 A. (a) What is the current in the second wire? (b) Are the two currents in the same direction or in opposite directions?

28.33 • **Lamp Cord Wires.** The wires in a household lamp cord are typically 3.0 mm apart center to center and carry equal currents in opposite directions. If the cord carries current to a 100-W light bulb connected across a 120-V potential difference, what force per meter does each wire of the cord exert on the other? Is the force attractive or repulsive? Is this force large enough so it should be considered in the design of the lamp cord? (Model the lamp cord as a very long straight wire.)

28.34 • A long, horizontal wire AB rests on the surface of a table and carries a current I . Horizontal wire CD is vertically above wire AB and is free to slide up and down on the two vertical metal guides C and D (Fig. E28.34). Wire CD is connected through the sliding contacts to another wire that also carries a current I , opposite in direction to the current in wire AB . The mass per unit length of the wire CD is λ . To what equilibrium height h will the wire CD rise, assuming that the magnetic force on it is due entirely to the current in the wire AB ?

Figure E28.34



Section 28.5 Magnetic Field of a Circular Current Loop

28.35 • **BIO Currents in the Brain.** The magnetic field around the head has been measured to be approximately 3.0×10^{-8} G. Although the currents that cause this field are quite complicated, we can get a rough estimate of their size by modeling them as a single circular current loop 16 cm (the width of a typical head) in diameter. What is the current needed to produce such a field at the center of the loop?

28.36 • Calculate the magnitude and direction of the magnetic field at point P due to the current in the semicircular section of wire shown in Fig. E28.36. (Hint: Does the current in the long, straight section of the wire produce any field at P ?)

Figure E28.36

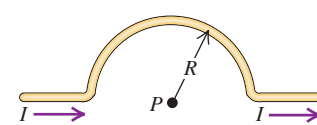
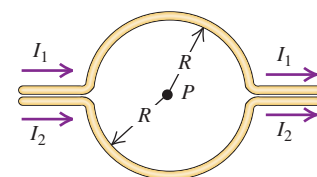


Figure E28.37



28.37 • Calculate the magnitude of the magnetic field at point P of Fig. E28.37 in terms of R , I_1 , and I_2 . What does your expression give when $I_1 = I_2$?

28.38 • A closely wound, circular coil with radius 2.40 cm has 800 turns. (a) What must the current in the coil be if the magnetic field at the center of the coil is 0.0580 T? (b) At what distance x from the center of the coil, on the axis of the coil, is the magnetic field half its value at the center?

28.39 • A closely wound, circular coil with a diameter of 4.00 cm has 600 turns and carries a current of 0.500 A. What is the magnitude of the magnetic field (a) at the center of the coil and (b) at a point on the axis of the coil 8.00 cm from its center?

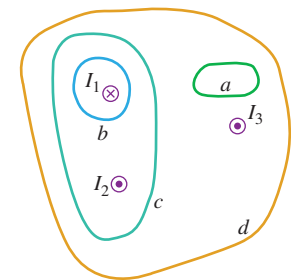
28.40 • A closely wound coil has a radius of 6.00 cm and carries a current of 2.50 A. How many turns must it have if, at a point on the coil axis 6.00 cm from the center of the coil, the magnetic field is 6.39×10^{-4} T?

28.41 • Two concentric circular loops of wire lie on a tabletop, one inside the other. The inner wire has a diameter of 20.0 cm and carries a clockwise current of 12.0 A, as viewed from above, and the outer wire has a diameter of 30.0 cm. What must be the magnitude and direction (as viewed from above) of the current in the outer wire so that the net magnetic field due to this combination of wires is zero at the common center of the wires?

Section 28.6 Ampere's Law

28.42 • Figure E28.42 shows, in cross section, several conductors that carry currents through the plane of the figure. The currents have the magnitudes $I_1 = 4.0$ A, $I_2 = 6.0$ A, and $I_3 = 2.0$ A, and the directions shown. Four paths, labeled a through d , are shown. What is the line integral $\oint \vec{B} \cdot d\vec{l}$ for each path? Each integral involves going around the path in the counterclockwise direction. Explain your answers.

Figure E28.42



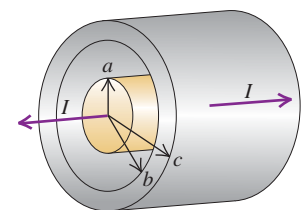
28.43 • A closed curve encircles several conductors. The line integral $\oint \vec{B} \cdot d\vec{l}$ around this curve is 3.83×10^{-4} T·m. (a) What is the net current in the conductors? (b) If you were to integrate around the curve in the opposite direction, what would be the value of the line integral? Explain.

Section 28.7 Applications of Ampere's Law

28.44 • As a new electrical technician, you are designing a large solenoid to produce a uniform 0.150-T magnetic field near the center of the solenoid. You have enough wire for 4000 circular turns. This solenoid must be 1.40 m long and 2.80 cm in diameter. What current will you need to produce the necessary field?

28.45 • **Coaxial Cable.** A solid conductor with radius a is supported by insulating disks on the axis of a conducting tube with inner radius b and outer radius c (Fig. E28.45). The central conductor and tube carry equal currents I in opposite directions. The currents are distributed uniformly over the cross sections of each conductor. Derive an expression for the magnitude of the magnetic field (a) at points outside the central, solid conductor but inside the tube ($a < r < b$) and (b) at points outside the tube ($r > c$).

Figure E28.45



28.46 • Repeat Exercise 28.45 for the case in which the current in the central, solid conductor is I_1 , the current in the tube is I_2 , and these currents are in the same direction rather than in opposite directions.

28.47 • A long, straight, cylindrical wire of radius R carries a current uniformly distributed over its cross section. At what locations is the magnetic field produced by this current equal to half of its largest value? Consider points inside and outside the wire.

28.48 • A 15.0-cm-long solenoid with radius 0.750 cm is closely wound with 600 turns of wire. The current in the windings is 8.00 A. Compute the magnetic field at a point near the center of the solenoid.

28.49 • A solenoid is designed to produce a magnetic field of 0.0270 T at its center. It has radius 1.40 cm and length 40.0 cm, and the wire can carry a maximum current of 12.0 A. (a) What minimum number of turns per unit length must the solenoid have? (b) What total length of wire is required?

28.50 • A toroidal solenoid has an inner radius of 12.0 cm and an outer radius of 15.0 cm. It carries a current of 1.50 A. How many equally spaced turns must it have so that it will produce a magnetic field of 3.75 mT at points within the coils 14.0 cm from its center?

28.51 • A magnetic field of 37.2 T has been achieved at the MIT Francis Bitter National Magnetic Laboratory. Find the current needed to achieve such a field (a) 2.00 cm from a long, straight wire; (b) at the center of a circular coil of radius 42.0 cm that has 100 turns; (c) near the center of a solenoid with radius 2.40 cm, length 32.0 cm, and 40,000 turns.

28.52 • A toroidal solenoid (see Example 28.10) has inner radius $r_1 = 15.0$ cm and outer radius $r_2 = 18.0$ cm. The solenoid has 250 turns and carries a current of 8.50 A. What is the magnitude of the magnetic field at the following distances from the center of the torus: (a) 12.0 cm; (b) 16.0 cm; (c) 20.0 cm?

28.53 • A wooden ring whose mean diameter is 14.0 cm is wound with a closely spaced toroidal winding of 600 turns. Compute the magnitude of the magnetic field at the center of the cross section of the windings when the current in the windings is 0.650 A.

Section 28.8 Magnetic Materials

28.54 • A toroidal solenoid with 400 turns of wire and a mean radius of 6.0 cm carries a current of 0.25 A. The relative permeability of the core is 80. (a) What is the magnetic field in the core? (b) What part of the magnetic field is due to atomic currents?

28.55 • A toroidal solenoid with 500 turns is wound on a ring with a mean radius of 2.90 cm. Find the current in the winding that is required to set up a magnetic field of 0.350 T in the ring (a) if the ring is made of annealed iron ($K_m = 1400$) and (b) if the ring is made of silicon steel ($K_m = 5200$).

28.56 • The current in the windings of a toroidal solenoid is 2.400 A. There are 500 turns, and the mean radius is 25.00 cm. The toroidal solenoid is filled with a magnetic material. The magnetic field inside the windings is found to be 1.940 T. Calculate (a) the relative permeability and (b) the magnetic susceptibility of the material that fills the toroid.

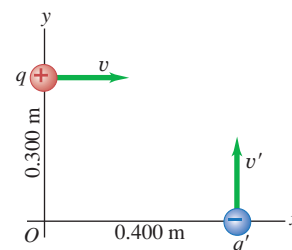
28.57 • A long solenoid with 60 turns of wire per centimeter carries a current of 0.15 A. The wire that makes up the solenoid is wrapped around a solid core of silicon steel ($K_m = 5200$). (The wire of the solenoid is jacketed with an insulator so that none of the current flows into the core.) (a) For a point inside the core, find the magnitudes of (i) the magnetic field $\vec{B}_0$ due to the solenoid current; (ii) the magnetization $\vec{M}$; (iii) the total magnetic field $\vec{B}$. (b) In a sketch of the solenoid and core, show the directions of the vectors $\vec{B}$, $\vec{B}_0$, and $\vec{M}$ inside the core.

28.58 • When a certain paramagnetic material is placed in an external magnetic field of 1.5000 T, the field inside the material is measured to be 1.5023 T. Find (a) the relative permeability and (b) the magnetic permeability of this material.

PROBLEMS

28.59 • A pair of point charges, $q = +8.00 \mu\text{C}$ and $q' = -5.00 \mu\text{C}$, are moving as shown in Fig. P28.59 with speeds $v = 9.00 \times 10^4 \text{ m/s}$ and $v' = 6.50 \times 10^4 \text{ m/s}$. When the charges are at the locations shown in the figure, what are the magnitude and direction of (a) the magnetic field produced at the origin and (b) the magnetic force that q' exerts on q ?

Figure P28.59



28.60 • At a particular instant, charge $q_1 = +4.80 \times 10^{-6} \text{ C}$ is at the point (0, 0.250 m, 0) and has velocity $\vec{v}_1 = (9.20 \times 10^5 \text{ m/s})\hat{i}$. Charge $q_2 = -2.90 \times 10^{-6} \text{ C}$ is at the point (0.150 m, 0, 0) and has velocity $\vec{v}_2 = (-5.30 \times 10^5 \text{ m/s})\hat{j}$. At this instant, what are the magnitude and direction of the magnetic force that q_1 exerts on q_2 ?

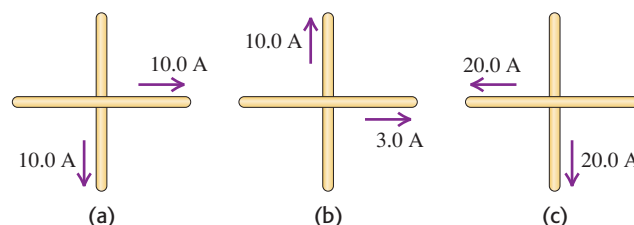
28.61 • Two long, parallel transmission lines, 40.0 cm apart, carry 25.0-A and 75.0-A currents. Find all locations where the net magnetic field of the two wires is zero if these currents are in (a) the same direction and (b) the opposite direction.

28.62 • A long, straight wire carries a current of 5.20 A. An electron is traveling in the vicinity of the wire. At the instant when the electron is 4.50 cm from the wire and traveling with a speed of $6.00 \times 10^4 \text{ m/s}$ directly toward the wire, what are the magnitude and direction (relative to the direction of the current) of the force that the magnetic field of the current exerts on the electron?

28.63 • CP A long, straight wire carries a 13.0-A current. An electron is fired parallel to this wire with a velocity of 250 km/s in the same direction as the current, 2.00 cm from the wire. (a) Find the magnitude and direction of the electron's initial acceleration. (b) What should be the magnitude and direction of a uniform electric field that will allow the electron to continue to travel parallel to the wire? (c) Is it necessary to include the effects of gravity? Justify your answer.

28.64 • Two very long, straight wires carry currents as shown in Fig. P28.64. For each case, find all locations where the net magnetic field is zero.

Figure P28.64



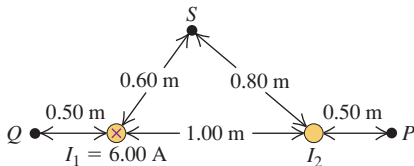
28.65 • CP Two identical circular, wire loops 40.0 cm in diameter each carry a current of 3.80 A in the same direction. These loops are parallel to each other and are 25.0 cm apart. Line ab is normal to the plane of the loops and passes through their centers.

A proton is fired at 2400 km/s perpendicular to line ab from a point midway between the centers of the loops. Find the magnitude of the magnetic force these loops exert on the proton just after it is fired.

28.66 • A negative point charge $q = -7.20$ mC is moving in a reference frame. When the point charge is at the origin, the magnetic field it produces at the point $x = 25.0$ cm, $y = 0$, $z = 0$ is $\vec{B} = (6.00 \mu\text{T})\hat{j}$, and its speed is 800 m/s. (a) What are the x -, y -, and z -components of the velocity $\vec{v}_0$ of the charge? (b) At this same instant, what is the magnitude of the magnetic field that the charge produces at the point $x = 0$, $y = 25.0$ cm, $z = 0$?

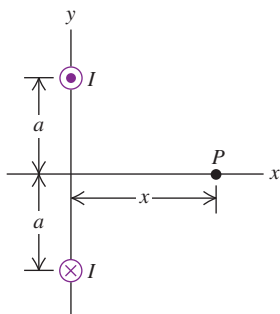
28.67 • Two long, straight, parallel wires are 1.00 m apart (Fig. P28.67). The wire on the left carries a current I_1 of 6.00 A into the plane of the paper. (a) What must the magnitude and direction of the current I_2 be for the net field at point P to be zero? (b) Then what are the magnitude and direction of the net field at Q ? (c) Then what is the magnitude of the net field at S ?

Figure P28.67



28.68 •• Figure P28.68 shows an end view of two long, parallel wires perpendicular to the xy -plane, each carrying a current I but in opposite directions. (a) Copy the diagram, and draw vectors to show the $\vec{B}$ field of each wire and the net $\vec{B}$ field at point P . (b) Derive the expression for the magnitude of $\vec{B}$ at any point on the x -axis in terms of the x -coordinate of the point. What is the direction of $\vec{B}$? (c) Graph the magnitude of $\vec{B}$ at points on the x -axis. (d) At what value of x is the magnitude of $\vec{B}$ a maximum? (e) What is the magnitude of $\vec{B}$ when $x \gg a$?

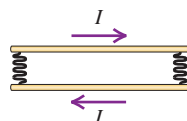
Figure P28.68



28.69 • Refer to the situation in Problem 28.68. Suppose that a third long, straight wire, parallel to the other two, passes through point P (see Fig. P28.68) and that each wire carries a current $I = 6.00$ A. Let $a = 40.0$ cm and $x = 60.0$ cm. Find the magnitude and direction of the force per unit length on the third wire, (a) if the current in it is directed into the plane of the figure, and (b) if the current in it is directed out of the plane of the figure.

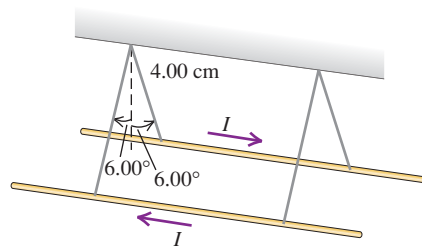
28.70 •• CP A pair of long, rigid metal rods, each of length L , lie parallel to each other on a perfectly smooth table. Their ends are connected by identical, very light conducting springs of force constant k (Fig. P28.70) and negligible unstretched length. If a current I runs through this circuit, the springs will stretch. At what separation will the rods remain at rest? Assume that k is large enough so that the separation of the rods will be much less than L .

Figure P28.70



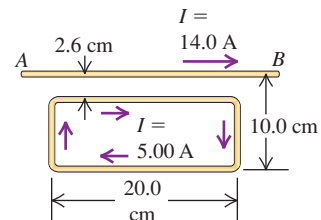
28.71 ••• CP Two long, parallel wires hang by 4.00-cm-long cords from a common axis (Fig. P28.71). The wires have a mass per unit length of 0.0125 kg/m and carry the same current in opposite directions. What is the current in each wire if the cords hang at an angle of 6.00° with the vertical?

Figure P28.71



28.72 • The long, straight wire AB shown in Fig. P28.72 carries a current of 14.0 A. The rectangular loop whose long edges are parallel to the wire carries a current of 5.00 A. Find the magnitude and direction of the net force exerted on the loop by the magnetic field of the wire.

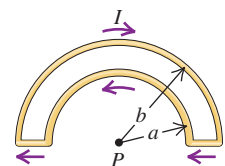
Figure P28.72



28.73 •• CP A flat, round iron ring 5.00 cm in diameter has a current running through it that produces a magnetic field of $75.4 \mu\text{T}$ at its center. This ring is placed in a uniform external magnetic field of 0.375 T. What is the maximum torque the external field can exert on the ring? Show how the ring should be oriented relative to the field for the torque to have its maximum value.

28.74 • The wire semicircles shown in Fig. P28.74 have radii a and b . Calculate the net magnetic field (magnitude and direction) that the current in the wires produces at point P .

Figure P28.74



28.75 • CALC Helmholtz Coils. Figure 28.75 is a sectional view of two circular coils with radius a , each wound with N turns of wire carrying a current I , circulating in the same direction in both coils. The coils are separated by a distance a equal to their radii. In this configuration the coils are called Helmholtz coils; they produce a very uniform magnetic field in the region between them. (a) Derive the expression for the magnitude B of the magnetic field at a point on the axis a distance x to the right of point P , which is midway between the coils. (b) Graph B versus x for $x = 0$ to $x = a/2$. Compare this graph to one for the magnetic field due to the right-hand coil alone. (c) From part (a), obtain an expression for the magnitude of the magnetic field at point P . (d) Calculate the magnitude of the magnetic field at P if $N = 300$ turns, $I = 6.00$ A, and $a = 8.00$ cm. (e) Calculate dB/dx and d^2B/dx^2 at $P(x = 0)$. Discuss how your results show that the field is very uniform in the vicinity of P .

Figure P28.75

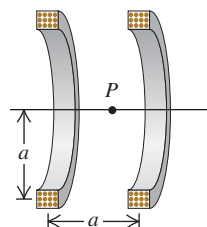
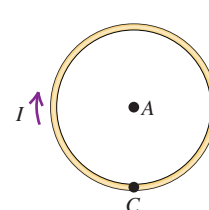


Figure P28.76



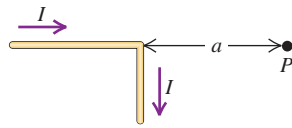
28.76 • A circular wire of diameter D lies on a horizontal table and carries a current I . In Fig. P28.76 point A marks the center of the circle and point C is on its rim. (a) Find the magnitude and direction of

the magnetic field at point A. (b) The wire is now unwrapped so it is straight, centered on point C, and perpendicular to the line AC, but the same current is maintained in it. Now find the magnetic field at point A. (c) Which field is greater: the one in part (a) or in part (b)? By what factor? Why is this result physically reasonable?

28.77 • CALC A long, straight wire with a circular cross section of radius R carries a current I . Assume that the current density is not constant across the cross section of the wire, but rather varies as $J = \alpha r$, where α is a constant. (a) By the requirement that J integrated over the cross section of the wire gives the total current I , calculate the constant α in terms of I and R . (b) Use Ampere's law to calculate the magnetic field $B(r)$ for (i) $r \leq R$ and (ii) $r \geq R$. Express your answers in terms of I .

28.78 • CALC The wire shown in Fig. P28.78 is infinitely long and carries a current I . Calculate the magnitude and direction of the magnetic field that this current produces at point P.

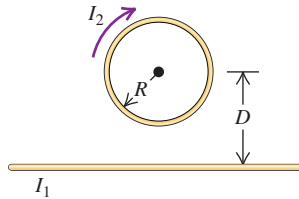
Figure P28.78



28.79 • A conductor is made in the form of a hollow cylinder with inner and outer radii a and b , respectively. It carries a current I uniformly distributed over its cross section. Derive expressions for the magnitude of the magnetic field in the regions (a) $r < a$; (b) $a < r < b$; (c) $r > b$.

28.80 • A circular loop has radius R and carries current I_2 in a clockwise direction (Fig. P28.80). The center of the loop is a distance D above a long, straight wire. What are the magnitude and direction of the current I_1 in the wire if the magnetic field at the center of the loop is zero?

Figure P28.80



28.81 • CALC A long, straight, solid cylinder, oriented with its axis in the z -direction, carries a current whose current density is $\vec{J}$. The current density, although symmetric about the cylinder axis, is not constant but varies according to the relationship

$$\vec{J} = \frac{2I_0}{\pi a^2} \left[1 - \left(\frac{r}{a} \right)^2 \right] \hat{k} \quad \text{for } r \leq a$$

$$= 0 \quad \text{for } r \geq a$$

where a is the radius of the cylinder, r is the radial distance from the cylinder axis, and I_0 is a constant having units of amperes. (a) Show that I_0 is the total current passing through the entire cross section of the wire. (b) Using Ampere's law, derive an expression for the magnitude of the magnetic field $\vec{B}$ in the region $r \geq a$. (c) Obtain an expression for the current I contained in a circular cross section of radius $r \leq a$ and centered at the cylinder axis. (d) Using Ampere's law, derive an expression for the magnitude of the magnetic field $\vec{B}$ in the region $r \leq a$. How do your results in parts (b) and (d) compare for $r = a$?

28.82 • A long, straight, solid cylinder, oriented with its axis in the z -direction, carries a current whose current density is $\vec{J}$. The current density, although symmetric about the cylinder axis, is not constant and varies according to the relationship

$$\vec{J} = \left(\frac{b}{r} \right) e^{(r-a)/\delta} \hat{k} \quad \text{for } r \leq a$$

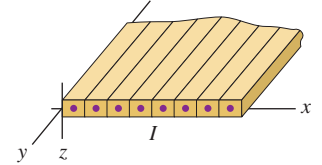
$$= 0 \quad \text{for } r \geq a$$

where the radius of the cylinder is $a = 5.00$ cm, r is the radial distance from the cylinder axis, b is a constant equal to 600 A/m, and δ is a constant equal to 2.50 cm. (a) Let I_0 be the total current passing

through the entire cross section of the wire. Obtain an expression for I_0 in terms of b , δ , and a . Evaluate your expression to obtain a numerical value for I_0 . (b) Using Ampere's law, derive an expression for the magnetic field $\vec{B}$ in the region $r \geq a$. Express your answer in terms of I_0 rather than b . (c) Obtain an expression for the current I contained in a circular cross section of radius $r \leq a$ and centered at the cylinder axis. Express your answer in terms of I_0 rather than b . (d) Using Ampere's law, derive an expression for the magnetic field $\vec{B}$ in the region $r \leq a$. (e) Evaluate the magnitude of the magnetic field at $r = \delta$, $r = a$, and $r = 2a$.

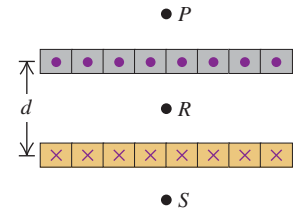
28.83 • An Infinite Current Sheet. Long, straight conductors with square cross sections and each carrying current I are laid side by side to form an infinite current sheet (Fig. P28.83). The conductors lie in the xy -plane, are parallel to the y -axis, and carry current in the $+y$ -direction. There are n conductors per unit length measured along the x -axis. (a) What are the magnitude and direction of the magnetic field a distance a below the current sheet? (b) What are the magnitude and direction of the magnetic field a distance a above the current sheet?

Figure P28.83



28.84 • Long, straight conductors with square cross section, each carrying current I , are laid side by side to form an infinite current sheet with current directed out of the plane of the page (Fig. P28.84). A second infinite current sheet is a distance d below the first and is parallel to it. The second sheet carries current into the plane of the page. Each sheet has n conductors per unit length. (Refer to Problem 28.83.) Calculate the magnitude and direction of the net magnetic field at (a) point P (above the upper sheet); (b) point R (midway between the two sheets); (c) point S (below the lower sheet).

Figure P28.84



28.85 • CP A piece of iron has magnetization $M = 6.50 \times 10^4$ A/m. Find the average magnetic dipole moment per atom in this piece of iron. Express your answer both in $\text{A} \cdot \text{m}^2$ and in Bohr magnetons. The density of iron is given in Table 14.1, and the atomic mass of iron (in grams per mole) is given in Appendix D. The chemical symbol for iron is Fe.

CHALLENGE PROBLEMS

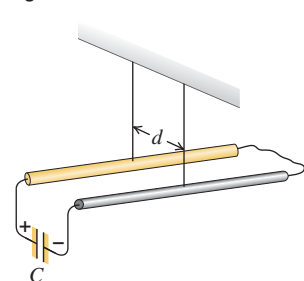
28.86 ••• A wide, long, insulating belt has a uniform positive charge per unit area σ on its upper surface. Rollers at each end move the belt to the right at a constant speed v . Calculate the magnitude and direction of the magnetic field produced by the moving belt at a point just above its surface. (Hint: At points near the surface and far from its edges or ends, the moving belt can be considered to be an infinite current sheet like that in Problem 28.83.)

28.87 ••• CP Two long, straight conducting wires with linear mass density λ are suspended from cords so that they are each horizontal, parallel to each other, and a distance d apart. The back ends of the wires are connected to each other by a slack, low-resistance connecting wire. A charged capacitor (capacitance C) is now added to the system; the positive plate of the capacitor (initial charge $+Q_0$) is connected to the front end of one of the wires, and the negative plate of the capacitor (initial charge $-Q_0$) is connected to the front end of the other wire (Fig. P28.87). Both of

these connections are also made by slack, low-resistance wires. When the connection is made, the wires are pushed aside by the repulsive force between the wires, and each wire has an initial horizontal velocity of magnitude v_0 . Assume that the time constant for the capacitor to discharge is negligible compared to the time it takes for any appreciable displacement in the position of the wires to occur. (a) Show that the initial speed v_0 of either wire is given by

$$v_0 = \frac{\mu_0 Q_0^2}{4\pi \lambda R C d}$$

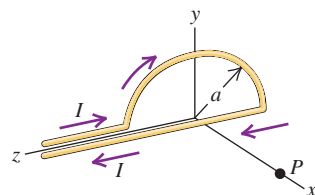
Figure P28.87



where R is the total resistance of the circuit. (b) To what height h will each wire rise as a result of the circuit connection?

28.88 ... CALC A wire in the shape of a semicircle with radius a is oriented in the yz -plane with its center of curvature at the origin (Fig. P28.88). If the current in the wire is I , calculate the magnetic-field components produced at point P , a distance x out along the x -axis. (Note: Do not forget the contribution from the straight wire at the bottom of the semicircle that runs from $z = -a$ to $z = +a$. You may use the fact that the fields of the two antiparallel currents at $z > a$ cancel, but you must explain *why* they cancel.)

Figure P28.88



Answers

Chapter Opening Question ?

There would be *no* change in the magnetic field strength. From Example 28.9 (Section 28.7), the field inside a solenoid has magnitude $B = \mu_0 n I$, where n is the number of turns of wire per unit length. Joining two solenoids end to end doubles both the number of turns and the length, so the number of turns per unit length is unchanged.

Test Your Understanding Questions

28.1 Answers: (a) (i), (b) (ii) The situation is the same as shown in Fig. 28.2 except that the upper proton has velocity $\vec{v}$ rather than $-\vec{v}$. The magnetic field due to the lower proton is the same as shown in Fig. 28.2, but the direction of the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ on the upper proton is reversed. Hence the magnetic force is attractive. Since the speed v is small compared to c , the magnetic force is much smaller in magnitude than the repulsive electric force and the net force is still repulsive.

28.2 Answer: (i) and (iii) (tie), (iv), (ii) From Eq. (28.5), the magnitude of the field dB due to a current element of length dl carrying current I is $dB = (\mu/4\pi)(I dl \sin \phi / r^2)$. In this expression r is the distance from the element to the field point, and ϕ is the angle between the direction of the current and a vector from the current element to the field point. All four points are the same distance $r = L$ from the current element, so the value of dB is proportional to the value of $\sin \phi$. For the four points the angle is (i) $\phi = 90^\circ$, (ii) $\phi = 0$, (iii) $\phi = 90^\circ$, and (iv) $\phi = 45^\circ$, so the values of $\sin \phi$ are (i) 1, (ii) 0, (iii) 1, and (iv) $1/\sqrt{2}$.

28.3 Answer: A This orientation will cause current to flow clockwise around the circuit. Hence current will flow south through the wire that lies under the compass. From the right-hand rule for the magnetic field produced by a long, straight, current-carrying conductor, this will produce a magnetic field that points to the left at the position of the compass (which lies atop the wire). The combination of the northward magnetic field of the earth and the westward field produced by the current gives a net magnetic field to the northwest, so the compass needle will swing counterclockwise to align with this field.

28.4 Answers: (a) (i), (b) (iii), (c) (ii), (d) (iii) Current flows in the same direction in adjacent turns of the coil, so the magnetic forces between these turns are attractive. Current flows in opposite directions on opposite sides of the same turn, so the magnetic forces between these sides are repulsive. Thus the magnetic forces on the

solenoid turns squeeze them together in the direction along its axis but push them apart radially. The *electric* forces are zero because the wire is electrically neutral, with as much positive charge as there is negative charge.

28.5 Answers: (a) (ii), (b) (v) The vector $d\vec{B}$ is in the direction of $d\vec{l} \times \vec{r}$. For a segment on the negative y -axis, $d\vec{l} = -\hat{k} dl$ points in the negative z -direction and $\vec{r} = x\hat{i} + a\hat{j}$. Hence $d\vec{l} \times \vec{r} = (a dl)\hat{i} - (x dl)\hat{j}$, which has a positive x -component, a negative y -component, and zero z -component. For a segment on the negative z -axis, $d\vec{l} = \hat{j} dl$ points in the positive y -direction and $\vec{r} = x\hat{i} + a\hat{k}$. Hence $d\vec{l} \times \vec{r} = 1(a dl)\hat{i} - 1(x dl)\hat{k}$, which has a positive x -component, zero y -component, and a negative z -component.

28.6 Answer: (ii) Imagine carrying out the integral $\oint \vec{B} \cdot d\vec{l}$ along an integration path that goes counterclockwise around the red magnetic field line. At each point along the path the magnetic field $\vec{B}$ and the infinitesimal segment $d\vec{l}$ are both tangent to the path, so $\vec{B} \cdot d\vec{l}$ is positive at each point and the integral $\oint \vec{B} \cdot d\vec{l}$ is likewise positive. It follows from Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$ and the right-hand rule that the integration path encloses a current directed out of the plane of the page. There are no currents in the empty space outside the magnet, so there must be currents inside the magnet (see Section 28.8).

28.7 Answer: (iii) By symmetry, any $\vec{B}$ field outside the cable must circulate around the cable, with circular field lines like those surrounding the solid cylindrical conductor in Fig. 28.20. Choose an integration path like the one shown in Fig. 28.20 with radius $r > R$, so that the path completely encloses the cable. As in Example 28.8, the integral $\oint \vec{B} \cdot d\vec{l}$ for this path has magnitude $B(2\pi r)$. From Ampere's law this is equal to $\mu_0 I_{\text{encl}}$. The net enclosed current I_{encl} is zero because it includes two currents of equal magnitude but opposite direction: one in the central wire and one in the hollow cylinder. Hence $B(2\pi r) = 0$, and so $B = 0$ for any value of r outside the cable. (The field is nonzero *inside* the cable; see Exercise 28.45.)

28.8 Answer: (i), (iv) Sodium and uranium are paramagnetic materials and hence are attracted to a magnet, while bismuth and lead are diamagnetic materials that are repelled by a magnet. (See Table 28.1.)

Bridging Problem

Answer: $B = \frac{\mu_0 n Q}{a}$